

# LMU MUNICH STATISTICS II COURSE NOTES

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## 25. EVENTS I

### Notations.

- $\Omega \neq \emptyset$  is called the *sample space* or *outcome space*.
  - A set  $A \subseteq \Omega$  is called an *event*.
  - $\omega_i \in \Omega = \{\omega_1, \omega_2, \dots\}$  is called an *outcome*.
  - $\{\omega_i\} \subseteq \Omega$  is called an *elementary event*.
- 

## 26. EVENTS II

### Example: Dice Roll. Outcomes

- 1: "Side with one dot is on top"
- ...
- 6: "Side with six dots is on top"

$$\Rightarrow \Omega = \{1, 2, 3, 4, 5, 6\}$$

### Some possible events

- $\{1\}$ : "Side with one dot is on top" (set with outcome 1)
  - $\{1, 2\}$ : "Side with one or two dots is on top"
  - $\{1, 3, 5\}$ : "Number of dots is odd"
  - $\{3, 4, 5, 6\}$ : "At least three dots"
  - $\Omega = \{1, 2, 3, 4, 5, 6\}$ : certain event
  - $\emptyset = \{\}$ : impossible event
-

## 27. POSSIBLE OUTCOMES

**Points on the unit circle:**

$$\Omega := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$


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## 28. EVENTS IV

**Possible Events.**

- Sets of individual points  $\{\omega \mid \omega \in \Omega\}$
- Open (closed, half-open) intervals  $(a, b) \subseteq \Omega$  ( $[a, b]$ ,  $(a, b]$ ,  $[a, b)$ ) on the circle
- Unions and intersections of intervals and individual points

**Remark.** For the outcome  $\omega_i \in \Omega$ , the set  $\{\omega_i\} \subseteq \Omega$  is an elementary event.

$$\underbrace{\omega_i \in \Omega}_{\text{Element}} \quad \underbrace{\{\omega_i\} \subseteq \Omega}_{\text{Subset}}$$


---

## 29. OPERATIONS ON EVENTS

**Equality:**  $A = B \iff \forall \omega \in \Omega : \omega \in A \iff \omega \in B$

**Subset:**  $A \subseteq B \iff \forall \omega \in \Omega : \omega \in A \Rightarrow \omega \in B$

**Intersection:**

$$A \cap B := \{\omega \in \Omega \mid (\omega \in A) \wedge (\omega \in B)\}$$

$$\bigcap_{i \in I} A_i := \{\omega \in \Omega \mid \forall i \in I : \omega \in A_i\}$$

**Union:**

$$A \cup B := \{\omega \in \Omega \mid (\omega \in A) \vee (\omega \in B)\}$$

$$\bigcup_{i \in I} A_i := \{\omega \in \Omega \mid \exists i \in I : \omega \in A_i\}$$

$$\biguplus_{i \in I} A_i := \{\omega \in \Omega \mid \exists i \in I : \omega \in A_i, A_i \cap A_j = \emptyset \text{ for } i \neq j\}$$

**Difference:**  $A \setminus B := \{\omega \in \Omega \mid (\omega \in A) \wedge (\omega \notin B)\}$

**Complement:**  $\bar{A} := \{\omega \in \Omega \mid \omega \notin A\}$

**Symmetric Difference:**

$$A \Delta B := (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$$

**Cardinality:**  $|A| := \text{Number of elements in } A$

**Cartesian Product:**

$$A \times B := \{(a, b) \mid a \in A, b \in B\}$$

$$\prod_{i \in I} A_i := \{(a_1, a_2, \dots, a_{|I|}) \mid a_i \in A_i, \forall i \in I\}$$

$$A^k = \prod_{i=1}^k A = \{(a_1, \dots, a_k) \mid a_i \in A, i = 1, \dots, k\}$$


---

## 30. INTERPRETATIONS

|                                 |                                                                |
|---------------------------------|----------------------------------------------------------------|
| $A \subseteq \Omega$            | $A$ occurs, is realized                                        |
| $\overline{A} \subseteq \Omega$ | $A$ does not occur, complement event                           |
| $A_1 \cup A_2$                  | $A_1$ or $A_2$ occurs                                          |
| $\bigcup_{i \in I} A_i$         | At least one of the $A_i$ occurs                               |
| $A_1 \cap A_2$                  | $A_1$ and $A_2$ occur                                          |
| $\bigcap_{i \in I} A_i$         | All $A_i$ occur                                                |
| $A_1 \cap A_2 = \emptyset$      | $A_1$ and $A_2$ do not occur simultaneously; they are disjoint |
| $A_1 \triangle A_2$             | Either $A_1$ or $A_2$ occurs                                   |
| $A_1 = A_2$                     | $A_1$ and $A_2$ describe the same event                        |
| $\Omega$                        | The certain event                                              |
| $\overline{\Omega} = \emptyset$ | The impossible event                                           |

---

## 31. LAWS AND PROPERTIES

Let  $A, B, C \subseteq \Omega$ .

**Reflexivity:**  $A \subseteq A$

**Antisymmetry:**  $A \subseteq B$  and  $B \subseteq A \Rightarrow A = B$

**Transitivity:**  $A \subseteq B$  and  $B \subseteq C \Rightarrow A \subseteq C$

$: \Rightarrow \mathcal{P}$  is partially ordered with respect to  $\subseteq$

**Commutative Laws:**

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

**Associative Laws:**

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

**De Morgan's Laws:**

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$$


---

## 33. GOAL OF THIS SECTION

- **Goal:** We want to define a general concept of probability.
  - **Approach:** Probability “measures” an event.
  - First, we define spaces of events (set systems) which we can then measure.
  - Then we define probability (measures) on these set systems: Every subset (event) is assigned a number between 0 and 1.
-



## 34. SET SYSTEMS

**Definition 2.4.**

- A *set* is a collection of different elements.
- A *set system* is a set of sets.
- $\mathcal{P}(\Omega) = \{A \mid A \subseteq \Omega\}$ , that is, the set of all subsets of the base set  $\Omega$ , is called the *power set*.
- An outcome of a random experiment is an element.
- $\Omega$  is the set of all possible outcomes.
- An event of a random experiment is a set of outcomes.
- A set of events is a set system.
- $\mathcal{P}(\Omega)$  is the set of all possible events.

**Convention for Notation.**

- $\mathcal{F}, \mathcal{A}, \mathcal{E}$  are set systems.
- $A, B$  are sets.

## 35. EXAMPLE: DICE ROLL

**Sets of Events (Set Systems).**

- $\{\{1\}\}$  (set containing the event  $\{1\}$ )
- $\{\{1, 3, 5\}, \{1, 2\}\}$
- $\{\Omega, \emptyset, \{1\}, \{2\}\}$

**Set of All Possible Events.** “Probability” must be defined for all possible events. Here, the following events are possible:

$$\begin{aligned} \mathcal{P}(\Omega) = & \{\emptyset, \{1\}, \{2\}, \dots, \{6\}, \\ & \{1, 2\}, \{1, 3\}, \dots, \{1, 6\}, \{2, 3\}, \dots, \{5, 6\}, \\ & \{1, 2, 3\}, \{1, 2, 4\}, \dots, \{4, 5, 6\}, \\ & \{1, 2, 3, 4\}, \dots, \{3, 4, 5, 6\}, \\ & \{1, 2, 3, 4, 5\}, \dots, \{2, 3, 4, 5, 6\}, \\ & \{1, 2, 3, 4, 5, 6\}\} \end{aligned}$$

## 36. EXAMPLE: SPINNING WHEEL I

- The set of all possible outcomes is uncountable.
- The set of all possible events consists of all countable and uncountable unions and intersections of events.

## 37. EXAMPLE: SPINNING WHEEL II

**Goal of the Definition of Probability.** Our goal is to define a function “probability”  $\mathbb{P}$  for as large a set of all possible events as possible. This function should satisfy the following meaningful properties:

- (1) **Normalization:**  $\mathbb{P}(\Omega) = 1$
- (2)  **$\sigma$ -additivity:** For any countable collection of pairwise disjoint sets  $A_1, A_2, \dots \subseteq \Omega$ , it holds:

$$\mathbb{P}(A_1 \cup A_2 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots$$

- (3) **Rotational Invariance:** The probability of any set  $A \subseteq \Omega$  does not change if the set is rotated by any angle.

### 38. VITALI'S THEOREM

#### Theorem 2.5. Theorem 2.5

A function  $\mathbb{P} : \mathcal{P}(\Omega) \rightarrow [0, 1]$ , which satisfies properties (1), (2), and (3), does not exist.

Proof will follow later.

### 40. $\sigma$ -ALGEBRA

Vitali's theorem shows that the power set  $\mathcal{P}(\Omega)$  may be too large (especially when  $\Omega$  is uncountable). We therefore restrict ourselves to smaller sets of *measurable* events.

**Definition 2.6 ( $\sigma$ -Algebra).** A set system  $\mathcal{F} \subseteq \mathcal{P}$  is called a  $\sigma$ -algebra over  $\Omega$ , or a closed set system, if:

- (S1):  $\Omega \in \mathcal{F}$  (Basic set)  
 (S2):  $A \in \mathcal{F} \Rightarrow \bar{A} \in \mathcal{F}$  (Complement)  
 (S3):  $A_i \in \mathcal{F}, i \in \mathbb{N} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$  (Countable union)

**Remark.** (S3) can also be defined in terms of intersections. Using (S2), we have:

$$\bigcup_{i=1}^{\infty} A_i = \overline{\bigcap_{i=1}^{\infty} \bar{A}_i}$$

### 41. EXAMPLES I

**Example 2.7 (Dice Roll).** With

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

we define the set system

$$\mathcal{F} = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$$

This is a  $\sigma$ -algebra.

- (S1)  $\Omega \in \mathcal{F}$
- (S2) For all  $A \in \mathcal{F}$ ,  $\bar{A} \in \mathcal{F}$
- (S3)  $\{1, 3, 5\} \cup \{2, 4, 6\} = \Omega$

**Example 2.8 (Smallest  $\sigma$ -Algebra).**

$$\{\emptyset, \Omega\}$$

### 42. EXAMPLES II

**Example 2.9 (Largest  $\sigma$ -Algebra).** The power set  $\mathcal{P}(\Omega)$  is a  $\sigma$ -algebra.

**But:** Vitali's theorem tells us that we cannot define a probability on all events in  $\mathcal{P}(\mathbb{R})$ !

Therefore, we need (especially for  $\Omega \subseteq \mathbb{R}$ ) a smaller set system that is still a  $\sigma$ -algebra.

### 43. MEASURABILITY

**Definition 2.10 (Measurable).** Let  $\mathcal{F}$  be a  $\sigma$ -algebra over  $\Omega$  and  $A \in \mathcal{F}$ . Then  $A$  is called *measurable* with respect to  $\mathcal{F}$ .

**Definition 2.11 (Measure Space).** Let  $\mathcal{F}$  be a  $\sigma$ -algebra over  $\Omega$ . The tuple  $(\Omega, \mathcal{F})$  is called a *measure space*.

---

#### 45. GENERATED $\sigma$ -ALGEBRA

**Definition 2.12 (Generated  $\sigma$ -Algebra).** If  $A \subseteq \Omega$ , then the set

$$\sigma(A) = \{\emptyset, A, \bar{A}, \Omega\}$$

is called the  $\sigma$ -algebra generated by  $A$ , and it is the smallest  $\sigma$ -algebra that contains  $A$ .

**Example 2.13.** In Example 2.7, we had:

$$\mathcal{F} = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$$

This was generated by  $\{1, 3, 5\}$ .

---

#### 46. INTERSECTION OF $\sigma$ -ALGEBRAS I

**Theorem 2.14.** Let  $I \neq \emptyset$  be an arbitrary index set and let  $\mathcal{F}_i$  be a  $\sigma$ -algebra over  $\Omega$  for all  $i \in I$ . Then

$$\mathcal{F} := \bigcap_{i \in I} \mathcal{F}_i$$

is a  $\sigma$ -algebra over  $\Omega$ .

**Proof:** According to Definition 2.6:

- (S2)  $A \in \mathcal{F} \Rightarrow A \in \mathcal{F}_i \forall i \in I \xrightarrow{(S2)} \bar{A} \in \mathcal{F}_i \forall i \in I \Rightarrow \bar{A} \in \mathcal{F}$
  - (S1) analogous
  - (S3) analogous
- 

#### 48. GENERATOR I

**Definition 2.16 (Generator).** Let  $\mathcal{E} \subseteq \mathcal{P}(\Omega)$  be a collection of subsets and let  $\Sigma$  be the set of all  $\sigma$ -algebras over  $\Omega$  that contain  $\mathcal{E}$ . Then the  $\sigma$ -algebra

$$\sigma(\mathcal{E}) := \bigcap_{\mathcal{F} \in \Sigma} \mathcal{F}$$

is called the  $\sigma$ -algebra generated by  $\mathcal{E}$ .

If for a  $\sigma$ -algebra  $\mathcal{A}$  it holds that

$$\sigma(\mathcal{E}) = \mathcal{A}$$

then  $\mathcal{E}$  is called a *generator* of  $\mathcal{A}$ .

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#### 51. BOREL $\sigma$ -ALGEBRA II

**Theorem 2.19 (Properties of  $\mathcal{B}$ ).** .

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#### 54. BOREL $\sigma$ -ALGEBRA IV

Definition of  $\mathcal{B}$  is also possible using closed or half-open intervals.  
 We will use these definitions synonymously in the following.  
 Sets not in  $\mathcal{B}$  but in  $\mathcal{P}(\mathbb{R})$  include, for example, the **Vitali sets**.

---

56. BOREL  $\sigma$ -ALGEBRA VII

We can also define the Borel  $\sigma$ -algebra on higher-dimensional spaces:

**Definition 0.1** (Definition 2.20). Let  $\mathcal{O}^n$  be the set of open intervals in  $\mathbb{R}^n$ , i.e.,

$$\mathcal{O}^n = \{U \subset \mathbb{R}^n \mid U \text{ open}\}.$$

Then the  $\sigma$ -algebra generated by  $\mathcal{O}^n$ , denoted by  $\mathcal{B}^n = \mathcal{B}(\mathbb{R}^n)$ , is called the Borel  $\sigma$ -algebra on  $\mathbb{R}^n$ .

Alternatively, we can restrict the space, e.g., let  $\mathcal{B}([0, 1])$  be the  $\sigma$ -algebra generated by

$$\mathcal{O}_{[0,1]} = \{[a, b[ \mid a, b \in \mathbb{R}, 0 \leq a < b \leq 1\}$$

## 61. MEASURE I

**Notation**

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$$

denotes the *extended real numbers*.

**Definition 0.2** (Definition 3.1 (Measure)). Let  $(\Omega, \mathcal{F})$  be a measurable space. A function

$$\mu : \mathcal{F} \rightarrow \overline{\mathbb{R}}$$

is called a *measure on  $\mathcal{F}$*  if the following conditions are satisfied:

(M1)  $\mu(\emptyset) = 0$

(M2)  $\mu(A) \geq 0 \quad \forall A \in \mathcal{F}$

(M3) For every sequence of disjoint sets  $(A_n)_{n \in \mathbb{N}} \in \mathcal{F}$ , the  $\sigma$ -additivity holds:

$$\mu \left( \bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n)$$

## 62. MEASURE II

- If there exists a sequence  $(A_n)$  of sets from  $\mathcal{F}$  such that

$$\bigcup_{n=1}^{\infty} A_n = \Omega \quad \text{and} \quad \mu(A_n) < \infty \quad \forall n \in \mathbb{N},$$

then  $\mu$  is called  *$\sigma$ -finite*.

- If  $\mu(\Omega) < \infty$ , then  $\mu$  is called *finite*. Finite  $\mu \Rightarrow \sigma$ -finite.
- If  $\mu(\Omega) = 1$ , then  $\mu$  is called a *probability measure*.

**Example.** Let  $\Omega$  be the set of students in the WTG lecture.

- Measures: Weight, cash in wallet (finite measures)
- Not a measure: Grade, gender

**Measure Space.**

**Definition 0.3** (Definition 3.2 (Measure Space)). If  $(\Omega, \mathcal{F})$  is a measurable space and  $\mu : \mathcal{F} \rightarrow \overline{\mathbb{R}}$  is a measure, then the triple  $(\Omega, \mathcal{F}, \mu)$  is called a *measure space*.

## 64. COUNTING MEASURE II

Compare Laplace probability:

$$p(A) = \frac{|A|}{|\Omega|} = \frac{\mu_Z(A)}{\mu_Z(\Omega)}$$


---

## 65. COUNTING MEASURE III. SKIP

**Theorem 0.1** (Theorem 3.4). *If  $\Omega$  is countable, then the counting measure is  $\sigma$ -finite.*

**Proof.** If  $\Omega$  is countable, then there exists an index set  $I \subseteq \mathbb{N}$  such that we can enumerate the elements of  $\Omega$  as:

$$\Omega = \{\omega_1, \omega_2, \dots\} = \bigcup_{i=1}^{\infty} \omega_i$$

Since  $\mu_Z(\{\omega_i\}) = 1 < \infty$ , the counting measure is finite on each singleton.  $\square$

- With respect to  $\mathbb{N}$ ,  $\mu_Z$  is not finite, but for example with respect to  $\{1, 2, 3, 4, 5, 6\}$  it is finite.
- 

## 66. COUNTING MEASURE IV

**Theorem 0.2** (Theorem 3.5). *Let  $A \in \mathcal{F}$  be fixed and at most countable. Then*

$$\mu_Z|_A(B) = |A \cap B|$$

*defines a  $\sigma$ -finite measure and is called the reduced counting measure.*

Example: Let  $\Omega = \mathbb{R}$  and  $A = \{1, 2, 3, 4, 5, 6\}$ .

---

## 67. MEASURE UNIQUENESS THEOREM I

**Definition 0.4** (Stable under Intersection). *A system of sets  $\mathcal{F} \subset \mathcal{P}$  is called stable under intersection if:*

$$A, B \in \mathcal{F} \implies A \cap B \in \mathcal{F}.$$

*Then  $\mathcal{F}$  is also called a  $\pi$ -system.*

Example:

- The collection of open intervals  $\mathcal{O}$  is stable under intersection.
- 

## 68. MEASURE UNIQUENESS THEOREM II

**Theorem 0.3** (Measure Uniqueness Theorem). *Let  $\mu$  and  $\nu$  be two measures on a measurable space  $(\Omega, \mathcal{F})$  and let  $\mathcal{E}$  be a generator of  $\mathcal{F}$  that is stable under intersection (a  $\pi$ -system), such that:*

- (1)  $\mu(E) = \nu(E) \quad \forall E \in \mathcal{E}$
- (2) *There exists a sequence  $(E_n) \subset \mathcal{E}$ ,  $n \in \mathbb{N}$ , of disjoint sets with*

$$\mu(E_n) = \nu(E_n) < \infty \quad \text{and} \quad \bigcup_{n=1}^{\infty} E_n = \Omega$$

*Then the two measures are identical:  $\mu = \nu$ .*

Corollary: If a measure is defined on a generator that is stable under intersection, it is uniquely determined.

---

### 69. LEBESGUE MEASURE I

**Definition 0.5** (Lebesgue Measure). *For the Lebesgue measure  $\lambda : \mathcal{B} \rightarrow \overline{\mathbb{R}}$ , the following holds for  $]a, b[ \in \mathcal{O}$ :*

$$\lambda(]a, b[) := b - a$$

The Lebesgue measure therefore assigns a length to an interval. For example, with  $\Omega = \mathbb{R}$ :

- $A = ]2, 4[ \Rightarrow \lambda(A) = 4 - 2 = 2$
  - $B = ]1, 8[ \Rightarrow \lambda(B) = 8 - 1 = 7$
  - $C = \{3\} \Rightarrow \lambda(C) = 0$
- 

### 70. LEBESGUE MEASURE II

- The Lebesgue measure can also be defined in higher dimensions:

**Definition 0.6** (Lebesgue Measure).

$$\lambda : \mathcal{B}^n \rightarrow \overline{\mathbb{R}}$$

With  $]a_i, b_i[ \in \mathcal{O} \ \forall i$ , we have:

$$\lambda(]a_1, b_1[ \times \cdots \times ]a_n, b_n[) = (b_1 - a_1) \cdots (b_n - a_n)$$

**Example:** Area of a rectangle.

---

### 71. LEBESGUE MEASURE III

- The Lebesgue measure is translation invariant. For  $(a, b) \in \mathcal{O}$ , we clearly have:

$$\lambda(]a + c, b + c[) = \lambda(]a, b[)$$

Therefore, more generally for  $A \in \mathcal{B}^n$  and  $c \in \mathbb{R}^n$ :

$$\lambda(A + c) = \lambda(A)$$

**Definition 0.7** (Reduced Lebesgue Measure). *The **reduced Lebesgue measure** (trace measure)  $\lambda|_A$  is defined as:*

$$\lambda_A(B) = \lambda(A \cap B)$$


---

### 72. INDICATOR FUNCTION I

**Definition 0.8** (Indicator Function, Definition 3.11). *Let  $A \subset \Omega$ . The indicator function  $I_A$  of  $A$  is the mapping  $I_A : \Omega \rightarrow \mathbb{R}$ , defined by:*

$$I_A(\omega) := \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

**Theorem 0.4** (Theorem 3.12). *Let  $A$  and  $B$  be two subsets of  $\Omega$ .*

- a)  $I_{\bar{A}} = 1 - I_A$
  - b)  $I_{A \cap B} = I_A \cdot I_B$
  - c) If  $A \cap B = \emptyset$ , then  $I_{A \cup B} = I_A + I_B$
  - d) If  $A \subset B$ , then  $I_{B \setminus A} = I_B - I_A$
-

## 73. INDICATOR FUNCTION II

**Example 0.1** (Example 3.13). Let  $A := \{1, 2, 3\}$  and  $B := \{1, 3, 5\}$ . Then:

- $I_A(2) = 1$
  - $I_B(2) = 0$
- 

## 74. DIRAC MEASURE I

**Definition 0.9** (Definition 3.14 (Dirac Measure)). Let  $\delta_\omega : \mathcal{F} \rightarrow \mathbb{R}$  with

$$\delta_\omega(A) := I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{otherwise} \end{cases}$$

**Example 0.2** (Example 3.15).

- For  $\mathcal{F} = \mathcal{P}(\mathbb{Z})$ :  $\delta_0(\{-1, 0, 1\}) = 1$
  - For  $\mathcal{F} = \mathcal{B}$ :  $\delta_0([0, 3]) = 0$
- 

## 75. DIRAC MEASURE II

**Definition 0.10** (Definition 3.16). Let  $\Omega$  be a set of outcomes. For an  $\omega \in \Omega$ , the distribution

$$\mathbb{P}(\omega) = \delta_\omega$$

is called the Dirac distribution or point mass distribution at  $\omega$ .

Thus, we have:

$$\mathbb{P}(\omega) = 1 \quad \text{and} \quad \mathbb{P}(\omega') = 0 \quad \text{for all } \omega' \neq \omega.$$

Definition is wrong. Should be

$$\mathbb{P}(A) = \delta_\omega(A)$$

Then  $\mathbb{P}(\{\omega\}) = 1$  and  $\mathbb{P}(\{\omega'\}) = 0$  for all  $\omega' \neq \omega$ .

---

## 77. PROBABILITY AS A NORMALIZED MEASURE I

- Let  $\Omega$  be a sample space.
- Potentially interesting events are subsets of  $\Omega$ , i.e., elements of a  $\sigma$ -algebra  $\mathcal{F}$  of the sample space.

**Definition 0.11** (Probability Measure). Let  $(\Omega, \mathcal{F})$  be a measurable space and

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

a measure on  $(\Omega, \mathcal{F})$  with  $\mathbb{P}(\Omega) = 1$ .

Then  $\mathbb{P}$  is called a probability measure and assigns each event  $A \in \mathcal{F}$  the probability  $\mathbb{P}(A)$ .  
 $\mathbb{P}$  is also called a distribution.

**Definition 0.12** (Probability Space).  $(\Omega, \mathcal{F}, \mathbb{P})$  is called a probability space.

---

## 78. EXAMPLE: DICE ROLL I

We roll a (six-sided) die. A reasonable sample space is

$$\Omega := \{1, 2, 3, 4, 5, 6\}.$$

- We are interested in the event  $A :=$  “the number rolled is even.”
- The  $\sigma$ -algebra generated by this event is  $\mathcal{F} = \sigma(A) = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$ .
- We follow the *principle of insufficient reason* (according to Laplace) and assume that all  $\omega \in \Omega$  are equally likely. This leads to the following probability measure  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ :
  - $\mathbb{P}(\{1, 3, 5\}) = 0.5$ ,
  - $\mathbb{P}(\{2, 4, 6\}) = 0.5$ .

Thus,  $(\Omega, \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}, \mathbb{P})$  is a probability space.

---

## 79. EXAMPLE: DICE ROLL II

**Definition 0.13** (Definition 3.19). *Let  $\Omega$  be finite and*

$$\mathbb{P} : \mathcal{P}(\Omega) \rightarrow [0, 1]$$

*with*

$$\mathbb{P}(\{\omega\}) := \frac{1}{|\Omega|}.$$

*We call  $\mathbb{P}(\{\omega\})$  the Laplace probability and  $\mathbb{P}$  the discrete uniform distribution on  $\Omega$ . We write*

$$U(\Omega) := \mathbb{P}.$$

Then  $(\Omega, \mathcal{P}(\Omega), U(\Omega))$  is a probability space.

---

## 81. CONTINUOUS UNIFORM DISTRIBUTION I

**Definition 0.14** (Definition 3.20). *Let  $\Omega = ]a, b[ \subset \mathbb{R}$ . Let  $a \leq c \leq d \leq b$ . Then*

$$\mathbb{P} : \mathcal{B}_{]a, b[} \rightarrow [0, 1]$$

*with*

$$\mathbb{P}(]c, d[) := \frac{d - c}{b - a}.$$

*We call  $\mathbb{P}$  the continuous uniform distribution on  $]a, b[$ . We write*

$$U(]a, b[) := \mathbb{P}.$$

Then  $(]a, b[, \mathcal{B}_{]a, b[, U(]a, b[))$  is a probability space.

---

## 82. ALMOST SURELY I

**Definition 0.15** (Definition 3.21 (Null Set)). *Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $A \in \mathcal{F}$  with  $\mu(A) = 0$ . Then  $A$  is called a  $(\mu)$ -null set.*

Examples.

- Individual points  $\{\omega \mid \omega \in \mathbb{R}\}$  are  $\lambda$ -null sets.
  - $\mathbb{N}$  and  $\mathbb{Q}$  are  $\lambda$ -null sets.
  - (Only)  $\emptyset$  is a  $\mu_Z$ -null set.
-



## 83. ALMOST SURELY II

**Definition 0.16** (Definition 3.22 ( $\mu$ -almost everywhere)). *Let the property  $E$  be meaningful for the elements  $\omega \in \Omega$  of a measure space  $(\Omega, \mathcal{F}, \mu)$ . We say that  $E$  holds ( $\mu$ -)almost everywhere if  $E$  holds for all  $\omega \notin N \subset \Omega$  and  $N$  is a  $\mu$ -null set.*

Example. The indicator function is  $\lambda$ -almost everywhere equal to zero.

Wrong. Should be: If  $A$  is a  $\lambda$ -null set (i.e., Lebesgue measure zero), then the indicator function  $I_A$  is  $\lambda$ -almost everywhere equal to 0.

**Definition 0.17** (Definition 3.23 ( $\mathbb{P}$ -almost surely)).

$$\mathbb{P}\text{-almost everywhere} \iff \mathbb{P}\text{-almost surely}$$

Broken Clock. A broken clock shows the correct time twice a day. The statement “The clock does not show the correct time” holds almost surely.

## 84. ALMOST SURELY III

**Example (Spinning Wheel)**

Let the event  $A =$  “The pointer lands exactly at the top.”

This event is a null set:

$$\mathbb{P}(A) = 0.$$

Therefore,  $A$  does not occur almost surely:

$A$  occurs  $\mathbb{P}$ -almost surely not.

## 85. CONDITIONAL PROBABILITY I

**Definition 3.24 (Conditional Probability)**

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $\mathbb{P}(B) > 0$  for some  $B \in \mathcal{F}$ . Then

$$\mathbb{P}(\cdot \mid B) : \mathcal{F} \rightarrow [0, 1]$$

with

$$A \mapsto \mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

is called the *conditional probability distribution* or simply the *conditional probability* of  $A$  given  $B$ .

## 86. CONDITIONAL PROBABILITY II

**Theorem 3.25**

$\mathbb{P}(\cdot \mid B)$  is a probability measure on  $(\Omega, \mathcal{F})$ .

**Proof**

M1) ✓

M2)  $\mathbb{P}(\cdot \mid B) \geq 0$  ✓

M3) Let  $A_i \in \mathcal{F}, i \in \mathbb{N}$ , disjoint. Show:

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) \stackrel{!}{=} \sum_{i=1}^{\infty} \mathbb{P}(A_i \mid B)$$

## 87. CONDITIONAL PROBABILITY III

$$\begin{aligned}
\sum_{i=1}^{\infty} \mathbb{P}(A_i \mid B) &= \frac{1}{\mathbb{P}(B)} \sum_{i=1}^{\infty} \mathbb{P}(A_i \cap B) \\
\mathbb{P} \text{ is a probability measure} &\implies \frac{1}{\mathbb{P}(B)} \mathbb{P}\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right) \\
&= \frac{1}{\mathbb{P}(B)} \mathbb{P}\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i \mid B\right)
\end{aligned}$$

**Normalization:**

$$\mathbb{P}(\Omega \mid B) = \frac{\mathbb{P}(\Omega \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1.$$


---

## 88. CONDITIONAL PROBABILITY IV

**Hypothetical Example.**

- $A$ : Passing the exam
- $B$ : Regular attendance in the lecture

$$\mathbb{P}(A) = 0.5 \quad \mathbb{P}(A \mid B) = 0.95$$


---

## 90. PROPERTIES OF THE MEASURE I

**Theorem 3.26 (Properties of the Measure).** Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $A, B, A_n \in \mathcal{F}$ ,  $n \in \mathbb{N}$ . Then the following hold:

- i) Finite additivity:  $A \cap B = \emptyset \implies \mu(A \cup B) = \mu(A) + \mu(B)$
- ii) Subadditivity:  $A \subset B$  and  $\mu(A) < \infty \implies \mu(B \setminus A) = \mu(B) - \mu(A)$
- iii) Monotonicity:  $A \subset B \implies \mu(A) \leq \mu(B)$
- iv) Sub- $\sigma$ -additivity:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

**Proof.**

- i) Consider the sequence  $A, B, \emptyset, \emptyset, \dots \xrightarrow{(M3)} \text{i)}$
- 

## 91. PROPERTIES OF THE MEASURE II

ii) and iii).

$$B = A \cup (B \setminus A) \xrightarrow{(M3)}$$

$$\mu(B) = \underbrace{\mu(A)}_{\geq 0} + \underbrace{\mu(B \setminus A)}_{\geq 0 \text{ by (M2)}} \geq \mu(A) \implies \text{(iii)}$$

$$\mu(B \setminus A) = \mu(B) - \mu(A) \iff \text{(ii) if } \mu(A) < \infty$$

iv).

$$\begin{aligned} \bigcup_{n=1}^{\infty} A_n &= \bigcup_{n=1}^{\infty} \left( A_n \setminus \bigcup_{k=1}^{n-1} A_k \right) \\ \Rightarrow \mu \left( \bigcup_{n=1}^{\infty} A_n \right) &\stackrel{(M3)}{=} \sum_{n=1}^{\infty} \mu \left( A_n \setminus \bigcup_{k=1}^{n-1} A_k \right) \stackrel{(ii)}{\leq} \sum_{n=1}^{\infty} \mu(A_n) \end{aligned}$$


---

## 92. CALCULATION RULES FOR PROBABILITIES

### Theorem 3.27 (Elementary Rules of Calculation)

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and  $A, B, A_n \in \mathcal{F}, n \in \mathbb{N}$ . Then the following hold:

- i)  $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$  (Complement rule)
- ii) If  $A \subset B$ , then  $\mathbb{P}(A) \leq \mathbb{P}(B)$  (Monotonicity)
- iii) **Inclusion-Exclusion Formula:**

$$\mathbb{P} \left( \bigcup_{i=1}^n A_i \right) = \sum_{i=1}^n (-1)^{i+1} \sum_{1 \leq j_1 < \dots < j_i \leq n} \mathbb{P} \left( \bigcap_{k=1}^i A_{j_k} \right)$$

- iv)  $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$  (Countable subadditivity)

### Proof.

Follows directly from Theorem 3.26. A more detailed proof of (iii) is given in the exercises.  $\square$

Especially for two events  $A$  and  $B$ :

- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
  - $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$
- 

## 93. CONVERGENCE OF MONOTONE SEQUENCES OF SETS

In the following: Let  $(A_n)_{n \in \mathbb{N}}$  be a sequence of sets (events).

- $A_n \uparrow A \iff A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots \text{ and } \bigcup_{i=1}^{\infty} A_i = A$
- $A_n \downarrow A \iff A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots \text{ and } \bigcap_{i=1}^{\infty} A_i = A$

### Example.

- $A_n := [0, 1 - \frac{1}{n}] \uparrow [0, 1[$
  - $B_n := [0, 1 + \frac{1}{n}] \downarrow [0, 1]$
- 

## 94. CONTINUITY OF MEASURE I

**Theorem 0.5.** *Theorem 3.28 (Continuity of Measure)* Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and let  $A, A_n \in \mathcal{F}, n \in \mathbb{N}$ . Then the following holds:

- i) *Continuity from below:*  $A_n \uparrow A \implies \mu(A_n) \uparrow \mu(A)$
- ii) *Continuity from above:*  $A_n \downarrow A \text{ and } \mu(A_1) < \infty \implies \mu(A_n) \downarrow \mu(A)$

This of course also holds for probability measures:

- Continuity from below:  $A_n \uparrow A \implies \mathbb{P}(A_n) \uparrow \mathbb{P}(A)$
  - Continuity from above:  $A_n \downarrow A \implies \mathbb{P}(A_n) \downarrow \mathbb{P}(A)$
-

## 95. CONTINUITY OF MEASURE II

**Proof**

i) Let  $A_0 := \emptyset$ .  $A_n \uparrow A$  means  $A_1 \subset A_2 \subset \dots$  and  $\bigcup_{n=1}^{\infty} A_n = A$ . Therefore,  $A = \bigcup_{n=1}^{\infty} A_n \setminus A_{n-1}$  is a disjoint union.

$$\Rightarrow \mu(A) \stackrel{(M3)}{=} \sum_{k=1}^{\infty} \mu(A_k \setminus A_{k-1}) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{k=1}^n A_k \setminus A_{k-1}\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$


---

## 96. CONTINUITY OF THE MEASURE III

ii).

$A \subset A_n \subset A_1$  by monotonicity  $\mu(A) < \infty$ ,  $\mu(A_n) < \infty \quad \forall n \in \mathbb{N}$   
 Since  $A_n \downarrow A$ , it follows that  $A_1 \setminus A_n \uparrow A_1 \setminus A$ , and thus also

$$\mu(A_1 \setminus A_n) \uparrow \mu(A_1 \setminus A).$$

Then:

$$\begin{aligned} \mu(A_1) - \mu(A) &= \mu(A_1 \setminus A) = \lim_{n \rightarrow \infty} \mu(A_1 \setminus A_n) \\ \lim_{n \rightarrow \infty} (\mu(A_1) - \mu(A_n)) &= \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_n) \\ &\Rightarrow \mu(A_n) \downarrow \mu(A) \end{aligned}$$


---

## 97. CONTINUITY OF THE MEASURE IV

**Example 3.29**

Let  $A_i := ]a, b + \frac{1}{i}[ \in \mathcal{O}$ . Then we have  $A_i \supset A_{i+1}$  and  $A_n \downarrow A = ]a, b]$ .  
 Hence also  $\lambda(A_i) \downarrow \lambda(A)$ , i.e.

$$\lambda(]a, b]) = \lambda(A) = \lim_{i \rightarrow \infty} \lambda(A_i) = b - a + \lim_{i \rightarrow \infty} \frac{1}{i} = b - a.$$

Analogous for closed intervals and points.

---

## 99. FLOOR FUNCTION

**Definition 0.18** (Definition 3.30). *The floor function (or Gauss bracket) is defined as*

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$$


---

## 100. VITALI SETS I

A **Vitali set**  $V$  is a subset of the interval  $(0, 1] \subset \mathbb{R}$  such that for every real number  $r \in \mathbb{R}$ , there exists exactly one  $v \in V$  such that  $(v - r) \in \mathbb{Q}$  is a rational number.

**Construction of Vitali Sets.**

- $\mathbb{Q} \subset \mathbb{R}$
  - We partition  $\mathbb{R}$  into disjoint “shifted copies” of  $\mathbb{Q}$  (i.e.,  $\mathbb{Q} + r$  for some  $r \in \mathbb{R}$ )
  - Each of these “shifted copies” has at least one element in  $[0, 1]$  (since  $r - \lfloor r \rfloor \in [0, 1]$  and  $\lfloor r \rfloor \in \mathbb{Q}$ )
  - We choose exactly one element in  $[0, 1]$  from each “shifted copy”. This is a Vitali set.
  - Every Vitali set is uncountable.
-

## 101. VITALI SETS II

We now show that  $\lambda(V)$  does not exist. To do this, we argue by contradiction, so we assume in the following that  $V$  is measurable.

- $\mathbb{Q}$  is countable. Let  $q_1, q_2, \dots$  be an enumeration of the rational numbers in  $[-1, 1]$ . The sets

$$V_k := \{v + q_k \mid v \in V\}, \quad k \in \mathbb{N}$$

are pairwise disjoint by the construction of the Vitali set.

- For all  $v \in V_k$  we have  $-1 \leq v \leq 2$ .
- For every  $r \in [0, 1]$ , there exists a rational number  $q_k \in [-1, 1]$  such that

$$r - v = q_k \quad \text{for some } k \in \mathbb{N}, \text{ i.e., } r \in V_k.$$

- Hence,

$$[0, 1] \subseteq \bigcup_{k=1}^{\infty} V_k \subseteq [-1, 2]$$

- By monotonicity of the measure, it follows that ...

## 102. VITALI SETS III

$$1 = \lambda([0, 1]) \leq \lambda\left(\bigcup_{k=1}^{\infty} V_k\right) \leq \lambda([-1, 2]) = 3$$

- Due to the translation invariance of the Lebesgue measure, we have for every  $k \in \mathbb{N}$ :  $\lambda(V_k) = \lambda(V)$ . Since the  $V_k$  are disjoint, by countable additivity:

$$\lambda\left(\bigcup_{k=1}^{\infty} V_k\right) = \sum_{k=1}^{\infty} \lambda(V_k) = \sum_{k=1}^{\infty} \lambda(V)$$

Hence:

$$1 \leq \sum_{k=1}^{\infty} \lambda(V) \leq 3$$

## 103. VITALI SETS IV

Now there are two possibilities:

- $\lambda(V) > 0$ , but this contradicts  $1 \leq \sum_{k=1}^{\infty} \lambda(V) \leq 3$ ,
- $\lambda(V) = 0$ , but this also contradicts  $1 \leq \sum_{k=1}^{\infty} \lambda(V) \leq 3$

Therefore, the Vitali set is not measurable!

## 107. MAPPING AND IMAGE

**Definition 0.19** (Function, Mapping). *A function or mapping  $f$  assigns to each element  $\omega_1$  from a domain  $\Omega_1$  exactly one element  $\omega_2$  from a codomain  $\Omega_2$ :*

$$f : \Omega_1 \rightarrow \Omega_2$$

$$f(\omega_1) = \omega_2 \quad \text{with } \omega_1 \in \Omega_1, \omega_2 \in \Omega_2$$

**Definition 0.20** (Image). *Let  $f : \Omega_1 \rightarrow \Omega_2$  be a mapping. Then the image of a set  $A \subset \Omega_1$  is defined by:*

$$f(A) := \{f(\omega) \in \Omega_2 \mid \omega \in A\}.$$

## 108. PREIMAGE I

**Definition 4.3 (Preimage)**

Let  $f : \Omega_1 \rightarrow \Omega_2$  be a mapping, then the *preimage* of a set  $B \subset \Omega_2$  is defined as

$$f^{-1}(B) := \{\omega \in \Omega_1 \mid f(\omega) \in B\}$$

Here,  $f$  does not need to be bijective, but it must preserve operations (*operation preservation*). For arbitrary sets  $B, B_i \in \Omega_2$ ,  $i \in I$ , the following holds:

$$\begin{aligned} f^{-1}\left(\bigcup_{i \in I} B_i\right) &= \left\{\omega \in \Omega_1 \mid f(\omega) \in \bigcup_{i \in I} B_i\right\} \\ &= \bigcup_{i \in I} \{\omega \in \Omega_1 \mid f(\omega) \in B_i\} \\ &= \bigcup_{i \in I} f^{-1}(B_i) \end{aligned}$$

## 109. PREIMAGE II

It holds:

$$\begin{aligned} f^{-1}\left(\bigcap_{i \in I} B_i\right) &= \bigcap_{i \in I} f^{-1}(B_i) \\ f^{-1}(\overline{B}) &= \{\omega \in \Omega_1 \mid f(\omega) \in \overline{B}\} \\ &= \Omega_1 \setminus \{\omega \in \Omega_1 \mid f(\omega) \in B\} \\ &= \Omega_1 \setminus f^{-1}(B) = f^{-1}(\overline{B}) \end{aligned}$$

**Remark.**

- $f^{-1}$  is called the **preimage function**.
- If  $f$  is bijective (this is exactly the case when  $f^{-1}(\omega_2)$  contains exactly one element  $\omega_1 \in \Omega_1$  for all elements  $\omega_2 \in \Omega_2$ ), then  $f^{-1}$  is also called the **inverse function**.

## 111. PREIMAGE OF A SYSTEM OF SETS I

**Definition 0.21** (Definition 4.4). Let  $f : \Omega_1 \rightarrow \Omega_2$  and  $\mathcal{F}_2 \subset \mathcal{P}(\Omega_2)$  be a system of sets. Then

$$f^{-1}(\mathcal{F}_2) := \{f^{-1}(B) \mid B \in \mathcal{F}_2\} \subset \mathcal{P}(\Omega_1)$$

**Theorem 0.6** (Theorem 4.5). Let  $\mathcal{F}_i$  be a  $\sigma$ -algebra over  $\Omega_i$ ,  $i = 1, 2$ . If  $f : \Omega_1 \rightarrow \Omega_2$  is a mapping, then  $f^{-1}(\mathcal{F}_2)$  is a  $\sigma$ -algebra over  $\Omega_1$ , and  $\{B \subset \Omega_2 \mid f^{-1}(B) \in \mathcal{F}_1\}$  is a  $\sigma$ -algebra over  $\Omega_2$ .

## 112. PREIMAGE OF A SYSTEM OF SETS II

**Proof.**

(S1), (S2), (S3) follow from the compatibility with operations, for example:

$$\begin{aligned}
\text{(S2)} \quad A \in f^{-1}(\mathcal{F}_2) &\implies A = f^{-1}(B), B \in \mathcal{F}_2 \\
&\implies \overline{B} \in \mathcal{F}_2 \\
&\implies \overline{A} = f^{-1}(B)^c = f^{-1}(\overline{B}) \in f^{-1}(\mathcal{F}_2)
\end{aligned}$$

□

**Example 0.3.** *Example 4.6*

- $\Omega_1 = \{1, \dots, 6\}^2$
- $f(\omega_1, \omega_2) = \omega_1 + \omega_2$
- $\Rightarrow \Omega_2 = \{2, \dots, 12\}$

Let  $\mathcal{F}_2 = \{\emptyset, \{7\}, \{2, \dots, 6, 8, \dots, 12\}, \Omega_2\}$  be a  $\sigma$ -algebra over  $\Omega_2$ .

$$\Rightarrow f^{-1}(\mathcal{F}_2) = \{\emptyset, A := \{(1, 6), (2, 5), \dots, (6, 1)\}, \overline{A}, \Omega_1\}$$

Let  $\mathcal{F}_1 = \{\emptyset, C = \{(1, 1), (2, 2), \dots, (6, 6)\}, \overline{C}, \Omega_1\}$  be a  $\sigma$ -algebra over  $\Omega_1$ .There exists no  $B \subset \Omega_2$  such that  $f^{-1}(B) = C$ .

## 114. MEASURABLE MAPPINGS I

**Definition 4.7 (Measurable Mapping)**Let  $(\Omega_1, \mathcal{F}_1)$  and  $(\Omega_2, \mathcal{F}_2)$  be two measurable spaces. A mapping  $f : \Omega_1 \rightarrow \Omega_2$  is called  $\mathcal{F}_1$ - $\mathcal{F}_2$ -measurable if

$$f^{-1}(\mathcal{F}_2) \subset \mathcal{F}_1.$$

In short, one speaks of a measurable mapping  $f$  if the  $\sigma$ -algebras involved are clearly known.

## 115. MEASURABLE MAPPINGS II

**Theorem 0.7** (Theorem 4.8). *Let  $(\Omega_i, \mathcal{F}_i), i = 1, 2$  be two measurable spaces, where  $\mathcal{F}_2 = \sigma(\mathcal{E})$  is generated by a system of sets  $\mathcal{E}$ .**Then a mapping  $f : \Omega_1 \rightarrow \Omega_2$  is  $\mathcal{F}_1$ - $\mathcal{F}_2$ -measurable if and only if*

$$f^{-1}(\mathcal{E}) \subset \mathcal{F}_1.$$

## 116. MEASURABLE MAPPINGS III

**Proof.**

$$\Rightarrow f \text{ is } \mathcal{F}_1\text{-}\mathcal{F}_2\text{-measurable}$$

$$\Rightarrow f^{-1}(\mathcal{F}_2) = \{f^{-1}(B) \mid B \in \mathcal{F}_2\} \subset \mathcal{F}_1$$

$$\mathcal{E} \subset \sigma(\mathcal{E}) = \mathcal{F}_2 \Rightarrow f^{-1}(\mathcal{E}) = \{f^{-1}(B) \mid B \in \mathcal{E}\} \subset \mathcal{F}_1$$

$$\Leftarrow f(\mathcal{F}_1) = \{B \subset \Omega_2 \mid f^{-1}(B) \in \mathcal{F}_1\} \text{ is a } \sigma\text{-algebra on } \Omega_2 \text{ (Theorem 4.5).}$$

$$f^{-1}(\mathcal{E}) \subset \mathcal{F}_1 \Rightarrow \mathcal{E} = f(f^{-1}(\mathcal{E})) \subset f(\mathcal{F}_1)$$

$f(\mathcal{F}_1)$  is thus a  $\sigma$ -algebra that contains  $\mathcal{E}$ .

$\Rightarrow \sigma(\mathcal{E}) \subset f(\mathcal{F}_1)$ , since  $\sigma(\mathcal{E})$  is the intersection of all  $\sigma$ -algebras containing  $\mathcal{E}$ .  $\square$

#### 117. MEASURABLE MAPPINGS IV

**Definition 0.22** (Example 4.9). Let  $(\Omega, \mathcal{F})$  be a measurable space. Then a mapping  $f : \Omega \rightarrow \mathbb{R}$  is  $\mathcal{F}$ - $\mathcal{B}$ -measurable if and only if

$$f^{-1}((-\infty, c]) = \{\omega \in \Omega \mid f(\omega) \leq c\} \in \mathcal{F} \quad \forall c \in \mathbb{R},$$

because  $(-\infty, c]$  forms a generating system of  $\mathcal{B}$  (follows from Theorem 2.19).

**Definition 0.23** (Example 4.10). For continuous mappings we have:

$$f : \mathbb{R} \rightarrow \mathbb{R} \text{ continuous} \iff f^{-1}(O) \in \mathcal{O} \quad \forall O \in \mathcal{O}$$

Here,  $\mathcal{O}$  is a generator of  $\mathcal{B} = \sigma(\mathcal{O})$  (cf. Def. 2.18), and hence  $f$  is  $\mathcal{B}$ - $\mathcal{B}$ -measurable.

#### 118. MEASURABLE FUNCTIONS V

##### Example 4.11

Let  $(\Omega, \mathcal{F})$  be a measurable space and  $A \subset \Omega$ .

We define the indicator function of the set  $A$  as:

$$I_A : \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto I_A(\omega) := \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$I_A^{-1} \in \{\emptyset, \Omega, A, \overline{A}\} \subset \mathcal{F} \iff A \in \mathcal{F}$$

This means that  $I_A$  is  $\mathcal{F}$ - $\mathcal{B}$ -measurable if and only if  $A \in \mathcal{F}$ .

Therefore,  $A$  is called *measurable* if  $A \in \mathcal{F}$  (see Definition 2.10).

#### 119. MEASURABLE MAPPINGS VI

##### Remarks.

- A function  $f : \Omega \rightarrow \mathbb{R}$  is called real-valued.
- A function  $f : \Omega \rightarrow \overline{\mathbb{R}}$  is called numerical.
- If  $(\Omega, \mathcal{F})$  is a measurable space, then  $f$  is called  $\mathcal{F}$ - $\overline{\mathcal{B}}$ -measurable if

$$\{\omega \in \Omega \mid f(\omega) \leq c\} \in \mathcal{F} \quad \text{for all } c \in \mathbb{R} \text{ with } \overline{\mathcal{B}} = \sigma(\mathcal{B} \cup \{\infty\} \cup \{-\infty\})$$

##### Notation.

$$\{f \leq c\} := \{\omega \in \Omega \mid f(\omega) \leq c\}$$

$$\{f < c\} := \{\omega \in \Omega \mid f(\omega) < c\}$$

$$\{f = c\} := \{\omega \in \Omega \mid f(\omega) = c\}$$



## 121. RANDOM VARIABLE DEFINITION

**Definition 4.12 (Random Variable)** Let  $(\Omega_1, \mathcal{F}_1, \mathbb{P})$  be a probability space and  $(\Omega_2, \mathcal{F}_2)$  a measurable space. Then a mapping

$$X : \Omega_1 \rightarrow \Omega_2$$

that is  $\mathcal{F}_1$ - $\mathcal{F}_2$ -measurable is called a *random variable (RV)*.

- If  $\Omega_2 = \mathbb{R}$ , then  $X$  is called a real-valued random variable.
  - If  $\Omega_2 = \overline{\mathbb{R}}$ , then  $X$  is called a numerical random variable.
  - If  $\Omega_2 = \mathbb{R}^n$ , then  $X$  is called an  $n$ -dimensional real-valued random variable.
- 

## 122. IMAGE MEASURE

**Definition 4.13 (Image Measure).** Let  $(\Omega_1, \mathcal{F}_1, \mu)$  be a measure space and  $(\Omega_2, \mathcal{F}_2)$  a measurable space, and let  $f : \Omega_1 \rightarrow \Omega_2$  be measurable. Then the function

$$\mu_f : \mathcal{F}_2 \rightarrow [0, \infty), \quad \mu_f(B) := \mu(f^{-1}(B))$$

is called the *image measure*  $\mu_f$  of  $\mu$  under  $f$ , and  $(\Omega_2, \mathcal{F}_2, \mu_f)$  is a measure space.

**Example 4.14.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto x + a$  be  $\mathcal{B}$ - $\mathcal{B}$ -measurable. Then:

$$\lambda_f(A) = \lambda(f^{-1}(A)) = \lambda(A - a) = \lambda(A) \quad \forall A \in \mathcal{B}$$

and  $\lambda_f$  is a measure on  $(\mathbb{R}, \mathcal{B})$ .

---

## 123. DISTRIBUTION OF A RANDOM VARIABLE

**Definition 0.24** (Definition 4.15 (Distribution of a Random Variable)). *Let  $(\Omega_1, \mathcal{F}_1, \mathbb{P})$  be a probability space and  $(\Omega_2, \mathcal{F}_2)$  a measurable space. Let  $X : \Omega_1 \rightarrow \Omega_2$  be a random variable. Then the image measure  $\mathbb{P}_X$  of  $\mathbb{P}$  under  $X$  is called the distribution of  $X$ .*

The image measure  $\mathbb{P}_X$  of  $\mathbb{P}$  under  $X$  is defined as

$$\mathbb{P}_X(A) = \mathbb{P}(X^{-1}(A)) \in [0, 1] \quad \forall A \in \mathcal{F}_2$$

where  $X^{-1}(A) \in \mathcal{F}_1$ .

It is again a probability measure.

---

## 124. WHEEL OF FORTUNE I

When spinning the wheel of fortune, you win:

- **100 Euros** if the pointer is in the **blue** region
  - **10 Euros** if the pointer is in the **green** region
  - **1 Euro** if the pointer is in the **red** region
-

## 125. WHEEL OF FORTUNE II

Let

$$\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

We can interpret the gain  $X$  as a random variable

$$X : \Omega \rightarrow \{1, 10, 100\}$$

Assuming the pointer position is uniformly distributed on  $\Omega$ , we get:

- $P_X(\{100\}) = P((x, y) \in \Omega \mid x > 0, y > 0) = \frac{1}{4}$
  - $P_X(\{10\}) = P((x, y) \in \Omega \mid x > 0, y < 0) = \frac{1}{4}$
  - $P_X(\{1\}) = P((x, y) \in \Omega \mid x < 0, y < 0) = \frac{1}{2}$
- 

## 126. BINOMIAL DISTRIBUTION I

**Example 4.16**

We consider the  $n$ -fold repetition of a random experiment with outcomes

$$\Omega_0 = \{\text{Failure}, \text{Success}\}.$$

Let  $p$  be the probability of success. Then the sample space is

$$\Omega = \prod_{i=1}^n \Omega_0.$$

The individual outcomes look, for example, like:

$$\omega = (\text{Success}, \text{Failure}, \text{Failure}, \text{Success}, \dots, \text{Success}).$$

Let  $k$  be the number of successes in  $\omega$ . Then:

$$\mathbb{P}(\{\omega\}) = p^k \cdot (1 - p)^{n-k}$$

Now let  $X$  be the measurable mapping, i.e., the random variable that assigns to each

$$\omega = (\omega_1, \dots, \omega_n)$$

the number of successes.

We are interested in the distribution of  $X$ . We have:

---

## 127. BINOMIAL DISTRIBUTION II

$$\begin{aligned} \mathbb{P}_X(\{k\}) &= \mathbb{P}(X^{-1}(\{k\})) = \sum_{\omega \in X^{-1}(\{k\})} \mathbb{P}(\{\omega\}) \\ &= \binom{n}{k} \cdot p^k (1 - p)^{n-k} \end{aligned}$$

**Definition 4.17**

The distribution  $\mathbb{P}_X$  in the example above is called the **binomial distribution**  $B(n, p)$ . The distribution  $B(1, p)$  is also called the Bernoulli distribution.

---

## 129. NOTATION

$$\begin{aligned}
\mathbb{P}_X(A) &= \mathbb{P}(X^{-1}(A)) \\
&= \mathbb{P}(\{\omega \in \Omega_1 \mid X(\omega) \in A\}) \\
&=: \mathbb{P}(X \in A)
\end{aligned}$$

$$\begin{aligned}
\mathbb{P}_X(\{c\}) &= \mathbb{P}(X^{-1}(\{c\})) \\
&= \mathbb{P}(\{\omega \in \Omega_1 \mid X(\omega) = c\}) \\
&=: \mathbb{P}(X = c)
\end{aligned}$$

( $\leq, \geq, <, >$  similarly.)

## 130. MEASURABILITY OF COMPOSITION

**Theorem 0.8** (Theorem 4.18 (Measurability of Composition)). *Let  $(\Omega_i, \mathcal{F}_i)$ ,  $i = 1, 2, 3$ , be measurable spaces and let  $f : \Omega_1 \rightarrow \Omega_2$  and  $g : \Omega_2 \rightarrow \Omega_3$  be measurable. Then the composition  $g \circ f : \Omega_1 \rightarrow \Omega_3$  is also measurable.*

**Proof**

$$\begin{aligned}
&f^{-1}(\mathcal{F}_2) \subset \mathcal{F}_1 \quad \text{and} \quad g^{-1}(\mathcal{F}_3) \subset \mathcal{F}_2 \\
\Rightarrow f^{-1}(\underbrace{g^{-1}(\mathcal{F}_3)}_{\subset \mathcal{F}_2}) &\subset \mathcal{F}_1 \Rightarrow (g \circ f)^{-1}(\mathcal{F}_3) \subset \mathcal{F}_1
\end{aligned}$$

□

**Remark 0.1.** *Remark Let  $(\Omega, \mathcal{F})$  be a measurable space and  $f = (f_1, \dots, f_n) : \Omega \rightarrow \mathbb{R}^n$ , and let  $\mathcal{B}^n$  be the Borel  $\sigma$ -algebra on  $\mathbb{R}^n$ . Then  $f$  is  $\mathcal{F}$ - $\mathcal{B}^n$ -measurable if and only if each  $f_i$ ,  $i = 1, \dots, n$ , is  $\mathcal{F}$ - $\mathcal{B}$ -measurable.*

*See Theorem 1.28 in Meintrup and Schäffler (2005).*

## 131. FUNCTIONS OF RANDOM VARIABLES

**Theorem 0.9** (Theorem 4.19 (Functions of Random Variables)). *Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random variable and let  $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be measurable. Then the composition  $g \circ X : \Omega \rightarrow \mathbb{R}^m$  is also a random variable.*

**Proof.**

The composition  $g \circ X$  is measurable as the composition of measurable functions ( $\mathcal{F}$ - $\mathcal{B}$ -measurable by Theorem 4.18), and hence a random variable. □

## 132. MEASURABLE COMBINATIONS I

**Theorem 0.10.** *Theorem 4.20 Let  $(\Omega, \mathcal{F})$  be a measurable space, and let  $f, g : \Omega \rightarrow \mathbb{R}$  be measurable functions and  $\alpha, \beta \in \mathbb{R}$ . Then the following functions are also measurable:*

- a)  $\alpha f + \beta g$
- b)  $f \cdot g$
- c)  $f/g$  if  $g(\omega) \neq 0$  for all  $\omega \in \Omega$

**Proof**

$$h : \Omega \rightarrow \mathbb{R}^2$$

$$\omega \mapsto h(\omega) = (f(\omega), g(\omega)) \quad \text{measurable (see remark to Theorem 4.18)}$$

$$\ell : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(x, y) \mapsto \ell(x, y)$$

**133. MEASURABLE COMBINATIONS II**

- a)  $\ell(x, y) = \alpha x + \beta y$  is continuous and thus measurable, so  $\ell \circ h = \alpha f + \beta g$  is measurable by Theorem 4.18.  
 b)  $\ell(x, y) = x \cdot y$  is measurable since it is continuous; same reasoning as in a).  
 c)  $\ell(x, y) = x/y$  is measurable since it is continuous; same reasoning as in a).  $\square$

Hence, we also have:

- If  $X$  is a random variable, then  $aX + b$  is a random variable ( $a, b \in \mathbb{R}$ )
- If  $X_1$  and  $X_2$  are random variables, then  $X_1 + X_2$  is a random variable (with  $X = (X_1, X_2)$  a random variable and  $g(X) = X_1 + X_2$ )
- Also,  $X_1 \cdot X_2$  and  $X_1/X_2$  are random variables
- $X^2$  is a random variable

**Example 0.4.** *Example 4.21 Let  $X$  be a random variable with  $\mathbb{P}_X = \mathcal{B}(n, p)$  (as in Example 4.16). Then  $n - X$  (the number of failures) is also a random variable.*

► Hence  $X + (n - X) = n$  is a random variable!

**134. MEASURABLE COMBINATIONS III****Notation**

$$\begin{aligned} \mathbb{P}_{g(X)}(A) &= \mathbb{P}((g \circ X)^{-1}(A)) \\ &= \mathbb{P}(\{\omega \in \Omega \mid (g \circ X)(\omega) \in A\}) \\ &:= \mathbb{P}(g(X) \in A) \end{aligned}$$

**Example 0.5.** *Example 4.22*

$$\mathbb{P}(X^2 \leq c) = \mathbb{P}(\{\omega \in \Omega \mid (X(\omega))^2 \leq c\})$$

**136. INDEPENDENCE OF EVENTS I**

**Definition 0.25** (Independence of Events, 4.23). *Events  $A_i \in \mathcal{F}$ ,  $i \in I \neq \emptyset$ , of a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  are called pairwise stochastically independent (stu) if for every finite subset  $\emptyset \neq J \subset I$  the following holds:*

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j).$$

In particular,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

for  $A, B \in \mathcal{F}$  stochastically independent.

Notation:  $A \perp B$

## 137. INDEPENDENCE OF EVENTS II

**Example 0.6** (4.24 — Fair Die).      •  $A$ : 'Even number'  $\rightarrow A := \{2, 4, 6\}$ ,  $\mathbb{P}(A) = \frac{1}{2}$ ,  
     •  $B$ : 'Number less than 3'  $\rightarrow B := \{1, 2\}$ ,  $\mathbb{P}(B) = \frac{1}{3}$ ,  
     •  $A \cap B = \{2\}$

$$\mathbb{P}(A \cap B) = \frac{1}{6} = \mathbb{P}(A) \cdot \mathbb{P}(B) \implies A \perp B$$

**Theorem 0.11** (4.25).      **a)**  $A, B$  are stochastically independent,  $\mathbb{P}(B) > 0 \iff \mathbb{P}(A | B) = \mathbb{P}(A)$

**b)**  $A, B$  are stochastically independent  $\iff A, \bar{B}$  are stochastically independent

*Proof.*      **a)**

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \stackrel{\text{stu}}{=} \frac{\mathbb{P}(A) \cdot \mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$$

□

## 138. INDEPENDENCE OF EVENTS III

*Continuation of part b).*

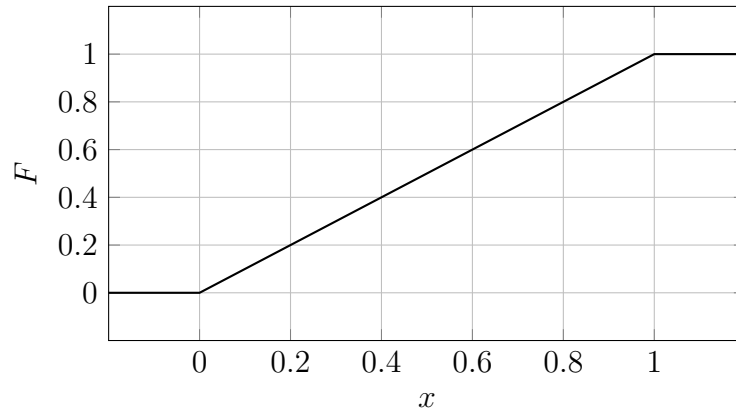
$$\begin{aligned} & \mathbb{P}(A) \cdot \mathbb{P}(B) = \mathbb{P}(A \cap B) \\ \iff & \mathbb{P}(A) \cdot (1 - \mathbb{P}(\bar{B})) = \mathbb{P}(A \cap B) \\ \iff & \mathbb{P}(A) - \mathbb{P}(A) \cdot \mathbb{P}(\bar{B}) = \mathbb{P}(A \cap B) \\ \iff & \mathbb{P}(A \cap B) + \mathbb{P}(A \cap \bar{B}) - \mathbb{P}(A) \cdot \mathbb{P}(\bar{B}) = \mathbb{P}(A \cap B) \\ \iff & \mathbb{P}(A \cap \bar{B}) = \mathbb{P}(A) \cdot \mathbb{P}(\bar{B}) \end{aligned}$$

□

## 152. EXAMPLE 5.3 (CONTINUOUS UNIFORM DISTRIBUTION)

*Idea:* Every (real) number in the interval  $[0, 1]$  is "equally likely." Thus,

$$F(x) = \mathbb{P}((-\infty, x]) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1 \end{cases}$$

FIGURE 1. Distribution function of  $U[0, 1]$ 

## 155. THEOREM 5.4 (PROPERTIES OF THE DISTRIBUTION FUNCTION)

A distribution function  $F_{\mathbb{P}}$  is:

- (1) monotonically increasing,
- (2) right-continuous,
- (3) and it holds that  $\lim_{x \rightarrow -\infty} F_{\mathbb{P}}(x) = 0$ ;  $\lim_{x \rightarrow \infty} F_{\mathbb{P}}(x) = 1$ .

## 156. PROPERTIES OF THE DISTRIBUTION FUNCTION II

**Proof.**

- i)  $a \leq b \Rightarrow F_{\mathbb{P}}(b) - F_{\mathbb{P}}(a) = \mathbb{P}((a, b]) \geq 0$
- ii)  $x_n \downarrow x \Rightarrow \lim_{n \rightarrow \infty} F_{\mathbb{P}}(x_n) = \lim_{n \rightarrow \infty} \mathbb{P}((-\infty, x_n]) = \mathbb{P}((-\infty, x]) = F_{\mathbb{P}}(x)$
- iii)  $x_n \downarrow -\infty \Rightarrow \lim_{n \rightarrow \infty} F_{\mathbb{P}}(x_n) = \lim_{n \rightarrow \infty} \mathbb{P}((-\infty, x_n]) = \mathbb{P}(\emptyset) = 0$   
 $x_n \uparrow \infty \Rightarrow \lim_{n \rightarrow \infty} F_{\mathbb{P}}(x_n) = \lim_{n \rightarrow \infty} \mathbb{P}((-\infty, x_n]) = \mathbb{P}(\mathbb{R}) = 1$

## 157. BENFORD'S LAW I

Newcomb (1835-1909) and later Frank Benford (1883-1948) made the surprising discovery that the leading digits 1-9 of whole numbers do not appear equally often in many cases. The leading digit 1 appears most frequently, followed by digit 2, and so on. Examples include:

- The frequency of leading digits in newspaper articles.
- The frequency of leading digits in tax documents.
- The frequency of the first digit of the file size of stored files.

Benford published the probability function for the random variable:

$$P(X = x) = \log_{10} \left( \frac{x+1}{x} \right) \cdot \mathbb{I}_{\{1, \dots, 9\}}(x)$$

Benford's Law is used, for example, in the detection of tax fraud or in verifying election results.

### 161. ALTERNATIVE DEFINITION

#### Definition 5.5 (Distribution Function)

Any monotonically increasing, right-continuous function

$$F : \mathbb{R} \rightarrow [0, 1] \text{ with } \lim_{x \rightarrow -\infty} F(x) = 0 \text{ and } \lim_{x \rightarrow \infty} F(x) = 1$$

is called a distribution function.

#### Questions

- Is every distribution completely determined by the distribution function?
- Does every distribution function correspond to a distribution?

### 162. CORRESPONDENCE THEOREM

#### Theorem 5.6 (Correspondence Theorem)

For every distribution function  $F$ , the measure  $\lambda_F([a, b]) = F(b) - F(a)$  is a probability measure with  $F_{\lambda_F} = F$ . Conversely, for every real probability measure  $P$ , there is a distribution function  $F$ .

- Probability measures on  $\mathbb{R}$  are therefore completely defined by their distribution functions.
- $\lambda_F$  is called the Lebesgue-Stieltjes measure of  $F$ .

### 163. CORRESPONDENCE THEOREM II

#### Proof.

- " $\implies$ ":  $F$  is a distribution function,  $\mu = \lambda_F$

$$F_\mu(x) = \lim_{n \rightarrow \infty} \mu([-n, x]) = \lim_{n \rightarrow \infty} (F(x) - F(-n)) = F(x) \quad \forall x \in \mathbb{R}$$

- " $\impliedby$ ": According to the first definition of the distribution function.

### 164. TRANSFORMATION OF THE DISTRIBUTION FUNCTION I

#### Example 5.7

Let  $\Omega = \{1, 2, \dots, n\}$  and  $X(\omega) = \omega$ . Then  $X$  is discretely uniformly distributed on  $\{1, \dots, n\}$  with the distribution function

$$F_X(x) = P(X \leq x) = \begin{cases} 0 & \text{for } x < 1, \\ \frac{\lfloor x \rfloor}{n} & \text{for } 1 \leq x < n, \\ 1 & \text{for } x \geq n. \end{cases}$$

Now let  $a \in \mathbb{N}$  and  $Y(\omega) = \omega + a - 1 = X(\omega) + a - 1$ . The distribution function of  $Y$  is

$$F_Y(y) = P(Y \leq y) = P(X + a - 1 \leq y) = P(X \leq y - a + 1) = F_X(y - a + 1).$$

## 165. TRANSFORMATION OF THE DISTRIBUTION FUNCTION II

**Theorem 5.8**

For invertible, strictly monotonically increasing transformations  $g$ , it holds that

$$F_{g(X)}(y) = F_X(g^{-1}(y))$$

**Proof.**

$$F_{g(X)}(y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

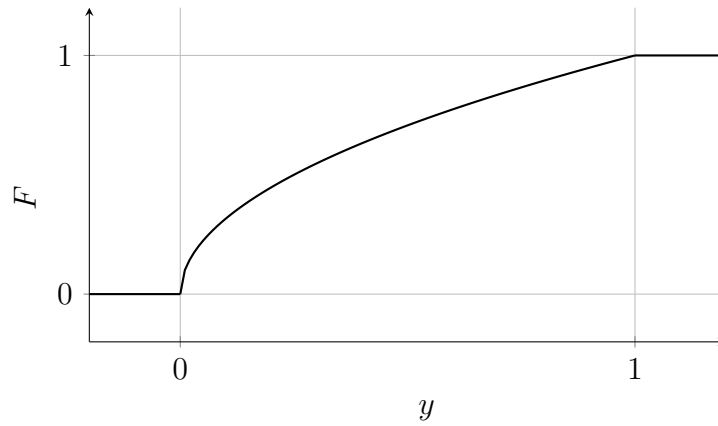

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## 166. TRANSFORMATION OF THE DISTRIBUTION FUNCTION III

**Example 5.9**

Let  $X \sim U(0, 1)$ .  $F_X(x) = x$  for  $0 \leq x \leq 1$ . What is the distribution function of  $Y = X^2$ ? Let  $g(x) = x^2$  which implies  $g^{-1}(y) = \sqrt{y}$ . For  $0 \leq x \leq 1$ , and hence  $0 \leq y \leq 1$ , it follows:

$$F_Y(y) = F_X(\sqrt{y}) = \sqrt{y}$$



## 168. MEDIAN

**Definition 5.10**

Given a one-dimensional random variable  $X$  with distribution  $\mathbb{P}_X$ . A real number  $m$  for which

$$\mathbb{P}_X((-\infty, m]) \geq 0.5 \quad \text{and} \quad \mathbb{P}_X([m, +\infty)) \geq 0.5$$

is called the **median** of  $\mathbb{P}_X$  or  $X$ .

**Example 5.11**

Let  $X \sim B(2, 0.5)$ , then  $m = 1$  is a median because  $\mathbb{P}(X \leq 1) = 0.75$  and  $\mathbb{P}(X \geq 1) = 0.75$ .

---

## 169. QUANTILE

**Definition 5.12**

Given a one-dimensional random variable  $X$  with distribution  $\mathbb{P}_X$ . A real number  $q$  for which

$$\mathbb{P}_X((-\infty, q]) \geq p \quad \text{and} \quad \mathbb{P}_X([q, +\infty)) \geq 1 - p$$

is called the  $p$ -quantile of  $\mathbb{P}_X$  or  $X$ .

---



## 171. DISCRETE PROBABILITY SPACE I

**Definition 6.1 (Discrete Probability Space)**

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space. If there exists a (finite or infinite) countable set  $T \in \mathcal{F}$  such that  $\mathbb{P}(\{\omega\}) > 0$  for all  $\omega \in T$  and  $\mathbb{P}(T) = 1$ , then

- $(\Omega, \mathcal{F}, \mathbb{P})$  is called a discrete probability space,
- $\mathbb{P}$  is called a discrete probability measure,
- $T$  is called a countable support of  $\mathbb{P}$ .

Note:  $\Omega$  itself does not need to be countable.

## 172. DISCRETE PROBABILITY SPACE II

- The function

$$f : \Omega \rightarrow [0, 1], \quad f(\omega) = \mathbb{P}(\{\omega\}) \cdot \mathbb{I}_T(\omega)$$

is called the counting density (more on this later). It holds that

$$\mathbb{P}(A) = \sum_{\omega \in A \cap T} f(\omega) \quad \forall A \in \mathcal{F}$$

## 173. MOMENTS

*As known from Statistics 1*

Let  $X$  be a discrete random variable with counting density  $f(x)$ . Then

- $\mathbb{E}(X) = \sum_T x f(x)$  is the expected value of  $X$  (a more general definition will follow later).
- $\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2)$  is the variance of  $X$ .

**Calculation Rules**

- $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$  (linearity, a more general proof will follow later).
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$ .

## 175. URN MODELS

It has become customary to describe certain basic situations that occur repeatedly in practical sampling as "urn models." One imagines an urn with balls and then draws a certain number of balls (sample) in a specific manner. In social science, each "ball" corresponds to a unit of interest (person, household), and the "urn" represents a hypothetical complete list of these units. A typical distinction in the various drawing processes is whether a unit can be included multiple times in a sample ("drawing with replacement," the drawn ball is returned to the urn) or "drawing without replacement" (the drawn balls remain outside the urn).

## 176. DRAWING WITH REPLACEMENT

- Population with  $N$  numbers  $G = \{1, \dots, N\}$ .
- Draw a sample of size  $n$  with replacement.
- To describe the random process, we need to determine the number of potentially possible sample outcomes (each sample is equally likely, Laplace assumption).
- $\Omega = \{(\omega_1, \dots, \omega_n) \mid \omega_j \in \{1, \dots, N\}\}$ , the same element can occur multiple times.
- $|\Omega| = \underbrace{N \cdot N \cdot \dots \cdot N}_{n \text{ times}} = N^n$ , i.e.,  $N^n$  potentially possible samples of size  $n$ .

## 177. DRAWING WITHOUT REPLACEMENT

- Population with  $N$  units (identified by numbers)  $G = \{1, \dots, N\}$ .
- Draw a sample of size  $n$  without replacement.
- $\Omega = \{(\omega_1, \dots, \omega_n) : \omega_j \in \{1, \dots, N\}, \omega_j \neq \omega_i \text{ for } i \neq j\}$ , each element can only occur once.
- Number of possible samples:

$$\begin{aligned}
 |\Omega| &= \underbrace{N}_{\text{1st draw}} \cdot \underbrace{(N-1) \cdots (N-n+1)}_{\text{2nd draw} \quad \text{---} \quad \text{n-th draw}} \\
 &= \frac{N \cdot (N-1) \cdots (N-n+1) \cdot (N-n) \cdots 2 \cdot 1}{(N-n) \cdot (N-n-1) \cdots 2 \cdot 1} \\
 &= \frac{N!}{(N-n)!}
 \end{aligned}$$


---

## 178. DRAWING WITHOUT REPLACEMENT WITHOUT CONSIDERING ORDER I

- Draw  $n$  balls from an urn with  $N$  numbered balls. The order of the draws does not matter, i.e., the sample “4,1,7” is not distinguished from “7,1,4”.
- $\Omega = \{\{\omega_1, \dots, \omega_n\} : \omega_j \in \{1, \dots, N\}, \omega_j \neq \omega_i \text{ for } j \neq i\}$

How large is the number of possible outcomes?

- First, consider the order. Then there are  $N!$  ways to arrange the balls. Since only the first  $n$  balls count, each  $(N-n)!$  arrangement that differs only from the  $(n+1)$ -th ball onward is equivalent. Therefore, we have

$$\frac{N!}{(N-n)!}$$

ways to generate ordered samples.

---

## 179. DRAWING WITHOUT REPLACEMENT WITHOUT CONSIDERING ORDER II

- Since the order among the first  $n$  balls does not matter, each  $n!$  ordered samples are equivalent. Therefore, the total number of different samples is

$$\frac{N!}{(N-n)!n!}$$

- Thus:

$$|\Omega| = \frac{N!}{(N-n)!n!} = \binom{N}{n}$$


---

## 180. DRAWING WITHOUT REPLACEMENT WITHOUT CONSIDERING ORDER III

**Definition 6.2**

The binomial coefficient

$$\binom{N}{n}$$

is defined as

$$\binom{N}{n} = \frac{N!}{(N-n)! \cdot n!}.$$


---

## 181. DRAWING WITHOUT REPLACEMENT WITHOUT CONSIDERING ORDER IV

The following holds:

- $\binom{N}{0} = 1$
  - $\binom{N}{1} = N$
  - $\binom{N}{N} = 1$
  - $\binom{N}{n} = 0$ , if  $N < n$
- 

## 182. EXAMPLE: LOTTO 6 OUT OF 49 I

- Question: How many different possible draws are there?
- Answer: Corresponds to drawing without replacement of 6 numbers out of 49, i.e.,

$$|\Omega| = \binom{49}{6} = \frac{49!}{43! \cdot 6!} = 13983816$$

Then it follows

$$P(\text{"6 correct"}) = \frac{1}{13983816} \approx 0.000000072$$


---

## 183. EXAMPLE: LOTTO 6 OUT OF 49 II

- Question: What is the probability of getting 5 correct numbers?
- Number of favorable cases:
  - Number of ways to choose 5 correct numbers out of 6:  $\binom{6}{5}$
  - Number of ways to choose 1 incorrect number out of 43:  $\binom{43}{1}$

$$P(\text{"5 correct"}) = \frac{\binom{6}{5} \cdot \binom{43}{1}}{\binom{49}{6}} = \frac{6 \cdot 43}{13983816} \approx 0.0000184$$


---

## 185. DISCRETE UNIFORM DISTRIBUTION I

$X \sim U(T)$  (see Statistics I)

- Also called Laplace distribution.

$$f(x) = \frac{1}{|T|} \mathbb{I}_T(x)$$

For  $X \sim U(1, 2, \dots, n)$ , it holds (see Example 5.7):

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{for } x < 1, \\ \frac{\lfloor x \rfloor}{n} & \text{for } 1 \leq x \leq n, \\ 1 & \text{for } x > n. \end{cases}$$


---

## 186. DISCRETE UNIFORM DISTRIBUTION II

$$\mathbb{E}(X) = \sum_{x=1}^n \frac{x}{n} = \frac{n(n+1)}{2n} = \frac{(n+1)}{2} \quad (\text{Gauss!})$$

$$\mathbb{E}(X^2) = \sum_{x=1}^n \frac{x^2}{n} = \frac{n(n+1)(2n+1)}{6n} = \frac{(n+1)(2n+1)}{6}$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{2^2} \\ &= \frac{8n^2 + 12n + 4 - (6n^2 + 12n + 6)}{24} = \frac{2n^2 - 2}{24} = \frac{n^2 - 1}{12} \end{aligned}$$

For example, for a die: The expected value is 3.5, and the variance is  $\frac{35}{12}$ .

---

## 187. DISCRETE UNIFORM DISTRIBUTION III

For  $Y \sim U(a, a+1, \dots, b)$ , with  $n = b - a + 1$ , it holds that  $Y$  and  $X + a - 1$  have the same distribution function and thus the same distribution, see Theorem 5.8. Therefore, it follows that

$$\mathbb{E}(Y) = \mathbb{E}(X + a - 1) = \mathbb{E}(X) + a - 1 = \frac{(b - a + 1) + 1}{2} + a - 1 = \frac{a + b}{2}$$

$$\text{Var}(Y) = \text{Var}(X + a - 1) = \text{Var}(X) = \frac{(b - a + 1)^2 - 1}{12}$$


---

## 188. GEOMETRIC DISTRIBUTION AND NEGATIVE BINOMIAL DISTRIBUTION I

$X \sim G(p)$  (See Statistics I)

- Event  $A$  occurs with probability  $p$
- $X$  is the number of trials until the first occurrence of  $A$  (discrete waiting time)
- $T = \mathbb{N}$ , Parameter  $0 \leq p \leq 1$

$$f(x) = (1 - p)^{x-1} p \cdot I_{\mathbb{N}}(x)$$

For  $X \sim G(p)$ , the following holds:

$$\mathbb{E}(X) = \frac{1}{p}$$

proof. For  $0 < p < 1$ , let  $q = 1 - p$ .

$$E(X) = \sum_{x=0}^{\infty} x q^{x-1} p = p \frac{1}{(1 - q)^2} = \frac{1}{p}$$

$$\text{Var}(X) = \frac{1 - p}{p^2}$$

**Memorylessness Property:**  $\mathbb{P}(X = x + x_0 \mid X > x_0) = \mathbb{P}(X = x)$

---

## 189. GEOMETRIC DISTRIBUTION AND NEGATIVE BINOMIAL DISTRIBUTION II

$X \sim NB(n, p)$  (See Statistics I)

- Event  $A$  occurs with probability  $\pi$
- $X$  is the number of trials until the  $n$ -th occurrence of  $A$
- $T = \{n, n+1, \dots\}$ , Parameter  $0 \leq \pi \leq 1$

$$f(x) = \binom{x-1}{n-1} p^n (1-p)^{x-n} I_T(x)$$

$$\mathbb{E}(X) = \frac{n}{\pi}$$

$$\text{Var}(X) = \frac{n(1-\pi)}{\pi^2}$$


---

## 190. HYPERGEOMETRIC DISTRIBUTION

$X \sim H(n, M, N)$  see Statistics I

*Recall.* The hypergeometric distribution describes the probability of obtaining a certain number of successes in a sample drawn without replacement from a finite population. Unlike the binomial distribution, where each trial is independent and has a fixed probability of success, the hypergeometric distribution deals with scenarios where the trials are dependent, meaning the outcome of one trial affects the probabilities of subsequent trials.

Suppose you have a population of size  $N$  which contains  $M$  successes (or "good" items) and  $N - M$  failures (or "bad" items). If you randomly draw a sample of size  $n$  from this population without replacement, the hypergeometric distribution gives the probability of obtaining exactly  $x$  successes in that sample.

- $T = \{0, 1, \dots, n\}$ , Parameter  $N \geq M \geq 0$

$$f(x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} I_T(x)$$


---

## 191. BERNOULLI AND BINOMIAL DISTRIBUTION I

- $X \sim B(1, p)$  Bernoulli distribution
- $X \sim B(n, p)$  Binomial distribution
- Number of successes in  $n$  trials with success probability  $p$
- $T = \{0, 1, \dots, n\}$ , Parameter  $0 \leq p \leq 1$

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} I_T(x)$$

**Theorem 6.3 (Relationship between  $B(1, p)$  and  $B(n, p)$ )**

Given  $X_1, \dots, X_n$  are stochastically independent with  $X_i \sim B(1, p)$ . Then:

$$X = \sum_{i=1}^n X_i \sim B(n, p)$$


---

## 192. BERNOULLI AND BINOMIAL DISTRIBUTION II

**Proof:**

$$\begin{aligned} \mathbb{P}(X_1 = 1, X_2 = 1, \dots, X_m = 1, X_{m+1} = 0, \dots, X_n = 0) \\ \stackrel{stu}{=} \prod_{i=1}^m \mathbb{P}(X_i = 1) \prod_{i=m+1}^n \mathbb{P}(X_i = 0) = \prod_{i=1}^m p \prod_{i=m+1}^n (1-p) = p^m (1-p)^{n-m} \end{aligned}$$

Since there are  $\binom{n}{m}$  ways to permute 1/0, it follows that

$$\mathbb{P}\left(\sum_{i=1}^n X_i = m\right) = \binom{n}{m} p^m (1-p)^{n-m} \implies \sum_{i=1}^n X_i \sim B(n, p)$$


---

## 193. BERNOULLI AND BINOMIAL DISTRIBUTION III

Sei  $X \sim B(1, p)$

Dann gilt

$$\begin{aligned} \mathbb{E}(X) &= \sum_{x \in \{0,1\}} x \cdot f_X(x) = 1 \cdot p + 0 \cdot (1-p) = p \\ \text{Var}(X) &= \sum_{x \in \{0,1\}} x^2 f_X(x) - p^2 \\ &= 1^2 \cdot p + 0^2 \cdot (1-p) - p^2 = p(1-p) \end{aligned}$$


---

## 194. BERNOULLI AND BINOMIAL DISTRIBUTION IV

Sei  $X \sim B(n, p)$

$$\begin{aligned} \mathbb{E}(X) &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} = n \sum_{x=0}^n \binom{n-1}{x-1} p^x (1-p)^{n-x} \\ &= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x} = np \sum_{x=0}^{n-1} \binom{n-1}{x} p^x (1-p)^{n-1-x} \\ &= np (p + (1-p))^{n-1} = np \end{aligned}$$


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## 195. BERNOULLI AND BINOMIAL DISTRIBUTION V

$$\begin{aligned} \text{Var}(X) &= \sum_{x=0}^n x^2 \binom{n}{x} p^x (1-p)^{n-x} - (np)^2 \\ &\quad q := 1-p \\ &= np \sum_{x=1}^n x \binom{n-1}{x-1} p^{x-1} q^{n-x} - (np)^2 \\ &= np \sum_{x=0}^{n-1} (x+1) \binom{n-1}{x} p^x q^{n-1-x} - (np)^2 \end{aligned}$$

$$\begin{aligned}
&= np \left( \sum_{x=0}^{n-1} x \binom{n-1}{x} p^x q^{n-1-x} + 1 \right) - (np)^2 \\
&= np((n-1)p + 1) - (np)^2 = np(1-p).
\end{aligned}$$


---

## 196. POISSON DISTRIBUTION I

$$X \sim Po(\lambda) \quad \text{see Statistics I}$$

- Distribution for count variable
- $T = \mathbb{N}_0$ , Parameter:  $\lambda \in \mathbb{R}^+$  (Rate)

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} I_T(x)$$


---

## 197. POISSON DISTRIBUTION II

**Example 6.4**

In one minute, an average of 2.5 cars drive inward towards the city through the Siegestor. Assuming that the cars drive independently of each other, the number of cars per hour follows a Poisson distribution with  $\lambda = 2.5$ . What is the probability of the event A: "Today, between 13:00:00 and 13:01:00, at most 2 cars pass by"?

$$\begin{aligned}
P(A) &= \sum_{x=0}^2 \frac{\lambda^x e^{-\lambda}}{x!} \cdot I_T(x) \\
&= \sum_{x=0}^2 \frac{2.5^x \cdot e^{-(2.5)}}{x!} \approx 0.5438
\end{aligned}$$


---

## 200. POISSON DISTRIBUTION V

**Theorem 6.5 (Relationship between Poisson and Binomial Distribution)**

Let  $p_n \in [0, 1]$ ,  $n \in \mathbb{N}$ , a sequence, and  $\lambda_n = n \cdot p_n \rightarrow \lambda > 0$  as  $n \rightarrow \infty$ . Then the following holds:

$$\binom{n}{k} p_n^k (1-p_n)^{n-k} \xrightarrow{n \rightarrow \infty} e^{-\lambda} \frac{\lambda^k}{k!} \quad \forall k \in \mathbb{N}.$$

Note.

$$\lim_{n \rightarrow \infty} \text{Binomial} \left( n, \frac{\lambda}{n} \right) = \text{Poisson}(\lambda)$$

This relationship shows that the Poisson distribution can be used to model the number of successes in a large number of Bernoulli trials, where the probability of success is very small.

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## EXTRA PROBLEM

**Example: Defects in Lightbulbs**

Suppose a factory produces lightbulbs. Each bulb has a probability of defect of  $p = \frac{\lambda}{n} = \frac{2}{1000} = 0.002$ . In one batch,  $n = 1000$  bulbs are produced. The expected number of defective bulbs is  $\lambda = n \cdot p = 1000 \cdot 0.002 = 2$ .

The number of defective bulbs  $X_n \sim \text{Binomial}(1000, 0.002)$ .

For large  $n$ , instead of computing binomial probabilities directly, we can approximate:  $X_n \approx \text{Poisson}(2)$ .

So for instance, the probability of finding exactly 3 defective bulbs is:

$$P(X = 3) \approx e^{-2} \frac{2^3}{3!} = e^{-2} \cdot \frac{8}{6} \approx 0.180.$$

Interpretation: Instead of using the complicated binomial formula with  $\binom{1000}{3}(0.002)^3(0.998)^{997}$ , we use the much simpler Poisson(2) formula. This is precisely the Poisson approximation to the binomial in action.

## 203. PROBABILITY GENERATING FUNCTION I

**Definition 0.26** (Probability Generating Function). For a random variable  $X : \Omega \rightarrow \mathbb{N}_0$ , the function  $m_X : [0, 1] \rightarrow [0, 1]$  is defined as

$$m_X(t) := \sum_{k=0}^{\infty} t^k P(X = k)$$

and is called the probability generating function.

**It holds that**

$$m_X(t) = E(t^X)$$

## 204. PROBABILITY GENERATING FUNCTION II

**For the probability generating function, it is a power series:**

$$P(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n$$

**Example 0.7** (Bernoulli Distribution). Let  $X \sim B(1, p)$ , thus  $\mathbb{P}(X = 0) = 1 - p$ ,  $\mathbb{P}(X = 1) = p$ , and  $\mathbb{P}(X = k) = 0$  for  $k \geq 2$ . Then we have:

$$m_X(t) = \sum_{k=0}^{\infty} t^k \mathbb{P}[X = k] = t^0(1 - p) + t^1 p = 1 - p + pt$$

## 205. PROPERTIES OF THE PROBABILITY GENERATING FUNCTION I

**Theorem 0.12** (Multiplication Formula). If  $X$  and  $Y$  are independent random variables taking values in  $\mathbb{N}_0$ , then we have

$$m_{X+Y}(t) = m_X(t)m_Y(t), \quad |t| \leq 1.$$

**Proof.** Skipped.



## 206. PROPERTIES OF THE PROBABILITY GENERATING FUNCTION II

**Example 0.8** (Example 6.9). Let  $X_i \sim B(1, p)$  for  $i = 1, \dots, n$  be stochastically independent and  $Y = \sum_i X_i \sim B(n, p)$ . The moment generating function of the binomial distribution is then:

$$m_Y(t) = \prod_{i=1}^n m_{X_i}(t) = (1 - p + pt)^n$$

## 207. MOMENT GENERATION I

If  $X : \Omega \rightarrow \mathbb{N}_0$ , then it holds that:

$$\mathbb{E}(X) = \lim_{t \uparrow 1} \frac{dm_X(t)}{dt}$$

More generally, for  $k \in \mathbb{N}$ , it holds that:

$$\mathbb{E}(X \cdot (X - 1) \cdots (X - k + 1)) = \lim_{t \uparrow 1} m_X^{(k)}(t)$$

## 208. MOMENT GENERATION II

**Example 0.9** (Example 6.10). Let  $X \sim B(n, p)$  with  $m_X(t) = (pt + 1 - p)^n$ :

$$m'_X(t) = np(pt + 1 - p)^{n-1}; \quad \mathbb{E}(X) = m'_X(1) = np$$

$$m''_X(t) = n(n-1)p^2(pt + 1 - p)^{n-2}; \quad m''_X(1) = n(n-1)p^2$$

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = m''_X(1) + m'_X(1) - m'_X(1)^2 = np(1 - p)$$

## 210. LEBESGUE INTEGRAL

- We know the Riemann integral. Here, we integrate over the  $x$ -axis of the real numbers (or over  $\mathbb{R}^d$ ).

**Goal:** We need a different type of integral for arbitrary measurable spaces. This allows us to define density, expectation, etc., in general.

- The **Lebesgue integral** is a very general concept of integration for arbitrary (!) measurable spaces and (Borel) measurable mappings into  $\mathbb{R}$ .

## 217. LEBESGUE INTEGRAL FOR INDICATOR FUNCTIONS I

In the following, let  $f(x) = I_A(x)$ .

Let

- $(\Omega, \mathcal{F}, \mu)$  be a measure space,
- $A$  a measurable set with respect to  $(\Omega, \mathcal{F})$ .

**Definition 0.27** (Definition 7.1 (Lebesgue Integral for Indicator Functions)). The Lebesgue integral for the indicator function  $I_A$  with respect to the measure  $\mu$  is defined as

$$\int_{\Omega} I_A d\mu := \int_{\Omega} I_A d\mu := \mu(A).$$

## 218. LEBESGUE INTEGRAL FOR INDICATOR FUNCTIONS II

**Example 0.10** (Example 7.2).  $\{\mathbb{R}, \mathcal{B}, \lambda\}; f(x) = I_{[-1,1]}(x)$

$$\int f d\lambda = \lambda([-1, 1]) = 2$$

$$\{\mathbb{N}, \mathcal{P}(\mathbb{N}), \mu_Z\}; f(x) = I_{\{-1,0,1\}}(x)$$

$$\int f d\mu_Z = \mu_Z(\{-1, 0, 1\}) = 3$$


---

## 219. LEBESGUE INTEGRAL FOR STEP FUNCTIONS I

**Definition 7.3 (Step Function)**

Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $f : \Omega \rightarrow \mathbb{R}$  an  $\mathcal{F}$ - $\mathcal{B}$ -measurable function with finite image

$$f(\Omega) = \{y_1, \dots, y_n\}.$$

Then there exist sets  $A_i \in \mathcal{F}$ ,  $i = 1, \dots, n$  such that  $f$  can be represented as

$$f = y_1 1_{A_1} + \dots + y_n 1_{A_n} = \sum_{i=1}^n y_i 1_{A_i},$$

where  $1_{A_i}$  denotes the indicator function of  $A_i$ .

The function  $f$  is then called a **step function** or **elementary function**.

- Set of step functions:

$$T := \{f \mid f : \Omega \rightarrow \mathbb{R}, f \text{ is a step function}\}$$

- Set of non-negative step functions:

$$T^+ := \{f \mid f \in T, f \geq 0\}$$

The definition works in general, but in the following we will always assume the normal representation, i.e. the sets  $A_i$  form a disjoint partition of  $\Omega$ .

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## 220. LEBESGUE INTEGRAL FOR STEP FUNCTIONS II

**Definition 7.4 (Lebesgue Integral for Step Functions)**

Let

$$f = \sum_{i=1}^n y_i 1_{A_i}, \quad \text{with } y_i \geq 0 \text{ for all } i,$$

be a non-negative step function. Then the Lebesgue integral of  $f$  with respect to  $\mu$  is defined as

$$\int f d\mu := y_1 \mu(A_1) + \dots + y_n \mu(A_n).$$


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## 221. LEBESGUE INTEGRAL FOR STEP FUNCTIONS III

**Example 0.11** (Example 7.5). Let  $\Omega = \{0, 1\}$  and  $(\Omega^2, \mathcal{P}(\Omega^2), \mathbb{P} = U(\Omega))$  be a probability space. Let  $X : \Omega^2 \rightarrow \mathbb{R}$  be a random variable with  $X(\omega) = \text{'Number of ones in } \omega\text{'}$ . Representation as a step function:

$$\begin{aligned} y_1 &= 0, A_1 = \{(0, 0)\}; \\ y_2 &= 1, A_2 = \{(0, 1), (1, 0)\}; \\ y_3 &= 2, A_3 = \{(1, 1)\}. \end{aligned}$$

It holds that

$$\int X d\mathbb{P} = 0 \cdot \mathbb{P}(A_1) + 1 \cdot \mathbb{P}(A_2) + 2 \cdot \mathbb{P}(A_3) = 1$$


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## 226. LEBESGUE INTEGRAL FOR NON-NEGATIVE MEASURABLE FUNCTIONS I

Let the following be defined:

- The set of measurable numeric functions  $M := \{f \mid f : \Omega \rightarrow \bar{\mathbb{R}}, f \text{ measurable}\}$ ,
- The set of non-negative measurable numeric functions  $M^+ := \{f \mid f \in M, f \geq 0\}$ .

**Lemma 0.1** (Lemma 7.7). If  $f \in M^+$  is a non-negative function, then there exists a sequence  $(f_n)_{n \in \mathbb{N}} \in T^+$  of non-negative step functions such that  $f_n \uparrow f$ .

---

## 235. LEBESGUE INTEGRAL WITH RESPECT TO THE DIRAC MEASURE I

**Theorem 7.12**

Let  $\omega \in \Omega$  and  $\delta_\omega$  be the Dirac measure. Let  $f$  be a measurable function. Then  $f$  is  $\delta_\omega$ -integrable if and only if  $f(\omega) < \infty$ , and in this case we have

$$\int f d\delta_\omega = f(\omega).$$


---

## 236. LEBESGUE INTEGRAL WITH RESPECT TO THE DIRAC MEASURE II

**Proof.**

First let  $f = \sum_i y_i 1_{A_i} \in T^+$ . Then  $\omega \in A_k$  for exactly one  $k$ . Thus

$$\int f d\delta_\omega = \sum_{i=1}^n y_i 1_{A_i} = y_k = f(\omega).$$

Now let  $f \in M^+$ . Then there exists a sequence  $f_n \uparrow f$  with  $f_n \in T^+$ . So we have

$$\int f d\delta_\omega = \lim_{n \rightarrow \infty} \int f_n d\delta_\omega = \lim_{n \rightarrow \infty} f_n(\omega) = f(\omega).$$

Finally, let  $f = f^+ - f^- \in M$ . Obviously,  $f$  is integrable if  $f^+(\omega) < \infty$  and  $f^-(\omega) < \infty$  (one of them is equal to zero). Thus it also follows that

$$\int f d\delta_\omega = f^+(\omega) - f^-(\omega) = f(\omega).$$


---

## 237. LEBESGUE INTEGRAL WITH RESPECT TO THE COUNTING MEASURE I

**Theorem 0.13** (Theorem 7.13). *Let  $(\mu_n)$  be a sequence of measures on  $\mathcal{F}$  and  $\mu = \sum_{i=1}^{\infty} \mu_i$  defined as*

$$\mu(A) = \sum_{i=1}^{\infty} \mu_i(A) \quad \forall A \in \mathcal{F}.$$

*For an  $\mathcal{F}$ -measurable function  $f$ , it holds that:*

$$f \text{ is } \mu\text{-integrable} \quad \Leftrightarrow \quad \sum_{i=1}^{\infty} \int |f| d\mu_i < \infty$$

*Then it holds that:*

$$\int f d\mu = \sum_{i=1}^{\infty} \int |f| d\mu_i.$$

## 238. LEBESGUE INTEGRAL WITH RESPECT TO THE COUNTING MEASURE II

*We will skip the exact proof. For convergence, one needs the \*\*big rearrangement theorem\*\* (If a series converges absolutely, the terms of the series can be grouped and rearranged arbitrarily).*

*However, the derivation of the integral follows the above procedure: step function, non-negative function over a sequence of step functions, arbitrary measurable function.*

## 239. LEBESGUE INTEGRAL WITH RESPECT TO THE COUNTING MEASURE III

**Theorem 7.14**

*Consider the measure space  $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mu_Z)$ . A function  $f : \mathbb{N} \rightarrow \mathbb{R}$  is  $\mu_Z$ -integrable if and only if the series  $\sum_{n=1}^{\infty} f(n)$  converges absolutely. In this case we have:*

$$\int_{\mathbb{N}} f d\mu_Z = \sum_{n=1}^{\infty} f(n).$$

*Thus, integration with respect to the counting measure is exactly summation!*

**Proof.**

*On  $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$  we have*

$$\mu_Z = \sum_{i=1}^{\infty} \delta_i.$$

*The rest follows from the previous theorem.*

□

## 241. MONOTONE CONVERGENCE THEOREM

**Theorem 0.14** (Monotone Convergence (MCT)). *Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and let  $(f_n)$  be a sequence of nonnegative measurable functions with  $f_n \uparrow f$  pointwise. Then*

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

**Example 0.12** (Classical increasing truncations). On  $([0, 1], \mathcal{B}, \lambda)$  define

$$f_n(x) := \min(nx, 1).$$

Then  $0 \leq f_1 \leq f_2 \leq \dots$  and  $f_n(x) \uparrow 1_{(0,1]}(x)$ . Moreover,

$$\int_0^1 f_n(x) dx = \int_0^{1/n} nx dx + \int_{1/n}^1 1 dx = 1 - \frac{1}{2n} \xrightarrow{n \rightarrow \infty} 1 = \int_0^1 1 dx.$$

Thus MCT justifies interchanging limit and integral.

## 242. DOMINATED CONVERGENCE THEOREM

**Theorem 0.15** (Dominated Convergence (DCT)). Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space. If  $f_n \rightarrow f$  pointwise a.e. and there exists  $g \in L^1(\mu)$  with  $|f_n| \leq g$  for all  $n$ , then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

**Example 0.13** (Powers  $x^n$  dominated by 1). On  $([0, 1], \mathcal{B}, \lambda)$  let  $f_n(x) := x^n$ . Then  $f_n(x) \rightarrow f(x)$  with

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1, \end{cases} \quad \text{so } f = 0 \text{ a.e.}$$

Since  $|f_n(x)| \leq 1 =: g(x)$  and  $g \in L^1([0, 1])$ , DCT applies:

$$\int_0^1 f_n(x) dx = \frac{1}{n+1} \xrightarrow{n \rightarrow \infty} 0 = \int_0^1 f(x) dx.$$

## 247. MOTIVATION

- A measure is a function on a  $\sigma$ -algebra  $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ , that is, on a (complicated) system of sets.
- We want to describe a measure more simply through a function on  $\Omega$  (the Radon-Nikodým density).
- Specifically, we want to represent a probability measure as a density relative to a measure.
- Question: Density relative to which measure?

Thanks to the unified theory developed up to this chapter, we can handle all conceivable distributions. Troublesome (and confusing) case distinctions (continuous, discrete, etc.) are no longer necessary.

## 249. DENSITY OF A MEASURE I

**Theorem 0.16** (Theorem 8.1). For every  $f \in M^+$ , the function

$$f \odot \mu : \mathcal{F} \rightarrow \bar{\mathbb{R}}$$

defined by

$$(f \odot \mu)(A) := \int_A f d\mu = \int f I_A d\mu$$

is a measure. Let  $\nu = f \odot \mu$ , then for all  $g \in M^+$ , it holds that

$$\int g d\nu = \int (gf) d\mu.$$

**Definition 0.28** (Definition 8.2 (Measure with Density, Density)). *Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $f \in M^+$ . Then  $f \odot \mu$  is called a measure with density  $f$  with respect to  $\mu$ . The function  $f : \Omega \rightarrow [0, \infty[$  is called the (Radon-Nikodým) density of the measure  $f \odot \mu$ .*

---

### 251. DENSITY OF A MEASURE III

**Example 8.3**

Let  $\mu = \mu_Z$  be the counting measure. Let

$$f(x) = \mathbb{P}(\{x\})$$

be the counting density of a discrete distribution  $\mathbb{P}$ . Then we have

$$(f \odot \mu_Z)(A) = \int_A f d\mu_Z = \int_A f 1_A d\mu_Z = \sum_{x \in A} f(x) \mu_Z(\{x\}) = \sum_{x \in A} f(x) = \mathbb{P}(A).$$

Thus, the counting density is a Radon-Nikodym density with respect to the counting measure.

---

### 252. DOMINATING MEASURE I

**Definition 8.4 (Absolute Continuity)**

Let  $\mu$  and  $\nu$  be two measures on  $(\Omega, \mathcal{F})$ . Then  $\nu$  is called absolutely continuous with respect to  $\mu$  if for all  $A \in \mathcal{F}$ , it holds that:

$$\mu(A) = 0 \implies \nu(A) = 0$$

In short:  $\nu \ll \mu$ . It is also said that  $\mu$  dominates  $\nu$ .

**Theorem 8.5**

For  $f$ , a measurable non-negative function ( $f \in M^+$ ), it holds that

$$\int f d\mu = 0 \iff \{\omega \in \Omega \mid f(\omega) > 0\} \text{ is a } \mu\text{-null set,}$$

where this set is called the support of  $f$ .

---

### 257. DENSITY OF A DISTRIBUTION

**Example 8.8**

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space. Let  $\mu$  be a measure that dominates  $\mathbb{P}$ , so that  $\mu(A) = 0 \implies \mathbb{P}(A) = 0$ . Then there exists a density  $f$  with  $\mathbb{P} = f \odot \mu$ , called the density of the distribution  $\mathbb{P}$ .

For  $\mathcal{F} = \mathcal{B}$ , the distribution function  $F(x)$  is given by

$$F(x) = \mathbb{P}((-\infty, x]) = \int_{(-\infty, x]} f d\mu$$

We distinguish:

- Discrete distributions, dominated by the counting measure  $\mu_Z$ :  $\int_A f d\mu_Z = \sum_{\omega \in A} f(\omega)$
  - (Absolutely) continuous distributions, dominated by the Lebesgue measure  $\lambda$ :  $\int_{[u, o]} f d\lambda = \int_u^o f(x) dx$ , where  $f$  is typically Riemann-integrable.
-

## 259. MIXED DISTRIBUTIONS I. SKIP!

**Example 8.9**

A smartphone manufacturer offers an extended repair warranty. From years of experience, they know that the proportion  $\alpha > 0$  of their smartphones will not require repairs during this period. The repair costs are assumed to be exponentially distributed with an expectation of  $\beta$  euros. Let  $X$  be the future repair costs of a newly sold smartphone. How is  $X$  distributed?

We construct a probability measure  $\mathbb{P}$  on  $(\mathbb{R}^+, \mathcal{B})$ . It holds that:

- $\mathbb{P}$  is absolutely continuous on  $\mathbb{R}^+$ ,
- $\mathbb{P}(0) = \alpha > 0$ .

## 260. MIXED DISTRIBUTIONS II. SKIP!

We are looking for the density and the dominating measure:

- $\mathbb{P}$  is not dominated by the Lebesgue measure:  $\lambda(\{0\}) = 0$ , but  $\mathbb{P}(0) > 0$ .
- $\mu_Z$  is not suitable since there is no  $\sigma$ -finite measure on  $(\mathbb{R}, \mathcal{B})$ .

Choose the measure  $\mu$  such that  $\mathbb{P} = f^* \odot \mu$ , a measure with density  $f^*$  with respect to  $\mu$ . Choose

$$\mu := \lambda + \delta_0.$$

$\mu$  is a  $\sigma$ -finite measure!

## 273. EXPECTATION I

**Definition 9.1 (Expectation)**

Let  $X : \Omega \rightarrow \mathbb{R}$  be a quasi-integrable real random variable, then the expectation of  $X$  is defined as

$$\mathbb{E}(X) := \int X d\mathbb{P} = \int x d\mathbb{P}_X(x)$$

The second equality follows from the following theorem:

**Theorem 9.2**

Let  $X : \Omega \rightarrow \mathbb{R}^n$  be an  $n$ -dimensional real random variable, and let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be a measurable function such that  $f \circ X : \Omega \rightarrow \mathbb{R}$  is integrable. Then it holds that

$$\int f d\mathbb{P}_X = \int f \circ X d\mathbb{P}.$$

## 280. EXAMPLES III

**Example 9.5**

We toss a fair coin twice.

- Let  $X = 1$  if the first toss is heads, and  $X = 0$  otherwise.
- Let  $Y$  be the number of heads in both tosses.

Obviously,  $X$  and  $Y$  are stochastically dependent! But

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y) = 0.5 + 1 = 1.5$$

## 281. ST. PETERSBURG PARADOX I

In a gambling game, for which an entry fee is charged, a fair coin is tossed until "heads" appears for the first time (geometric distribution).

As a prize, you receive  $2^{x-1}$  euros for  $x$  tosses. How much would you pay to participate in this game?

Let  $X$  be the random variable "number of tosses". The probability that "heads" appears for the first time on the  $x$ -th toss is

$$\mathbb{P}(X = x) = \left(\frac{1}{2}\right)^x$$


---

## 282. ST. PETERSBURG PARADOX I

How large is the expected payoff?

$$\mathbb{E}(2^{X-1}) = \sum_{k=1}^{\infty} \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 4 + \cdots = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot 2^{k-1} = \sum_{k=1}^{\infty} \frac{1}{2} = \infty.$$

Thus, the expected payoff is infinitely large, so would one, purely rationally speaking, accept an infinite stake??

---

## 285. VARIANCE I

**Definition 9.6 (Variance of a Random Variable)**

If  $X$  is a random variable with  $\mathbb{E}(|X|) < \infty$ , then the variance of  $X$  is defined as

$$\text{Var}(X) := \mathbb{E}((X - \mathbb{E}(X))^2)$$

Variance of  $X$ .

**Special cases** (under conditions similar to the expectation):

- Continuous random variable:  $\text{Var}(X) = \int (x - \mathbb{E}(X))^2 f_X(x) dx$
  - Discrete random variable:  $\text{Var}(X) = \sum_{x \in T} (x - \mathbb{E}(X))^2 f_X(x)$
- 

## 287. COVARIANCE AND CORRELATION

**Definition 9.8**

Let  $X$  and  $Y : \Omega \rightarrow \mathbb{R}$  be random variables. Then

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))]$$

is called the **covariance**. Furthermore, if  $\text{Var}(X) > 0$  and  $\text{Var}(Y) > 0$ , then

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

is called the **correlation** of  $X$  and  $Y$ .

- Covariance and correlation are measures of the strength and direction of the **linear** stochastic dependence between random variables.
-



## 298. MOMENTS II

**Definition 9.14**

If  $X^3$  is quasi-integrable, then the skewness of  $X$  is defined as

$$\gamma(X) := \frac{m_3^0(X)}{(m_2^0(X))^{\frac{3}{2}}} = \frac{\mathbb{E}((X - \mathbb{E}(X))^3)}{\sqrt{\text{Var}(X)^3}}$$

If  $X^4$  is quasi-integrable, then the kurtosis (or excess kurtosis) of  $X$  is defined as

$$\kappa(X) := \frac{m_4^0(X)}{(m_2^0(X))^2}$$

where this is the kurtosis or curvature of  $X$ .

**Interpretation:**

- Skewness = 0  $\rightarrow$  symmetric distribution (e.g. normal).
- Positive skewness  $\rightarrow$  distribution is right-skewed: long right tail, bulk of values on the left.
- Negative skewness  $\rightarrow$  distribution is left-skewed: long left tail, bulk of values on the right.

**Examples:**

- Log-normal distribution  $\rightarrow$  positively skewed.
- Exponential distribution  $\rightarrow$  positively skewed.

**Definition:** Kurtosis measures the “tailedness” or peakness of a distribution compared to the normal distribution. It’s about how much probability mass is in the tails and near the center.

**Examples:**

- Normal distribution  $\rightarrow$  mesokurtic.
- Exponential distribution  $\rightarrow$  leptokurtic.

**Visual intuition:**

- Skewness = direction of asymmetry.
- Kurtosis = thickness of tails + sharpness of peak.

## 299. SKEWNESS EXAMPLE I

**Example 9.15**

The log-normal distribution is therefore right-skewed (positively skewed), i.e. steep on the left.

## 308. MODE I

**Definition 9.19**

Let  $X$  be a one-dimensional random variable with distribution  $\mathbb{P}_X$  and density  $f(x)$ . A number  $m$  is called the mode of  $X$  or  $\mathbb{P}$  if  $f(m)$  is a local maximum of  $f(x)$ . That is, if the following holds:

$$f(x) \leq f(m) \text{ for all } x \in (m - \epsilon, m + \epsilon) \text{ for some } \epsilon > 0.$$

- The definition can easily be extended to multidimensional distributions.
- Since the densities of continuous random variables are almost surely unique, modes are also almost surely unique in a formal sense.

## 309. MODE II

**Examples**

- Let  $X \sim \text{Exp}(1)$  with density

$$f(x) = \lambda e^{-\lambda x} \mathbf{1}_{\mathbb{R}_0^+}(x).$$

The mode of  $X$  is 0 and  $\mathbb{P}(X > 0) = 1$ .

- The mode of the Binomial( $n, 0.5$ ) distribution is  $\frac{n}{2}$  when  $n$  is even. If  $n$  is odd, then  $\frac{n-1}{2}$  and  $\frac{n+1}{2}$  are both modes.

## 310. MODE III

**Definition 9.20**

If a continuous density has only one mode, the distribution is called *unimodal*. If a continuous density has several modes, the distribution is called *multimodal*; in the special case of two modes, *bimodal*.

## 313. LOCATION PARAMETERS I

As location parameters of a distribution, we refer to all parameters that describe the location of the density on the  $x$ -axis, for example:

- Expectation (mean)
- Mode
- Median

Only for unimodal symmetric distributions, the mean equals the median, which equals the mode — otherwise, this is not the case!

**Example 9.22**

For  $X \sim \text{LN}(\mu, \sigma^2)$  (log-normal, right-skewed), it holds that:

- $\mathbb{E}(X) = \exp(\mu + \frac{1}{2}\sigma^2)$
- $\text{Median}(X) = \exp(\mu)$
- $\text{Mode}(X) = \exp(\mu - \sigma^2)$

## 314. LOCATION PARAMETERS II

**For all right-skewed distributions:**

$$\mathbb{E}(X) \geq \text{Median}(X) \geq \text{Mode}(X)$$

**For all left-skewed distributions:**

$$\mathbb{E}(X) \leq \text{Median}(X) \leq \text{Mode}(X)$$

## 315. LOCATION PARAMETERS III

In many standard distributions, the mean  $\mathbb{E}(X)$  is fixed by symmetry (e.g.  $\mathbb{E}(X) = 0$  for the standard normal, central  $t$ , or central  $\chi^2$ ).

When you shift the distribution away from this “central” mean, you introduce a noncentrality parameter.

This parameter captures the extent to which the distribution deviates from the “central” (symmetric, zero-mean) case.

## 316. SCALE AND SHAPE PARAMETERS I

On the other hand, scale parameters (*Skalenparameter*) refer to all parameters that describe the width of the density, for example:

- Variance
- Interquartile range (75%-quantile minus 25%-quantile)

Shape parameters are parameters that describe the shape of the density.

## 317. SCALE AND SHAPE PARAMETERS II

**Definition 9.23**

Let  $X : \Omega \rightarrow \mathbb{R}$  be a random variable with continuous density

$$f_X(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx) \cdot I_{(0,\infty)}(x)$$

is called Gamma-distributed:  $X \sim \text{Ga}(a, b)$  with parameters  $a, b > 0$ .

- $a$  is called the shape parameter
- $b$  is called the inverse scale parameter

## 321. MARKOV AND CHEBYSHEV INEQUALITIES

**Example:** If variance is  $\sigma^2 = 4$ , then

$$\mathbb{P}(|X - \mu| \geq 6) \leq \frac{4}{36} = \frac{1}{9}.$$

So at least  $8/9$  of the probability mass lies within 6 units of the mean.

**Example:** On average, a bus arrives every 10 minutes. What is the maximum probability (by Markov's inequality) that someone has to wait 30 minutes or more?

$$\mathbb{P}(T \geq 30) \leq \frac{10}{30} = \frac{1}{3}.$$

Even without knowing the distribution of bus arrival times, we can say the chance of waiting  $\geq 30$  minutes is at most 33%.

**Example:** Suppose widget weights have mean 100g and standard deviation 2g. What percentage of widgets are guaranteed (by Chebyshev's inequality) to lie between 90g and 110g?

$$\mathbb{P}(|X - 100| \geq 10) \leq \frac{4}{100} = 0.04$$

So at least 0.96 of the probability mass lies within 10 units of the mean.

**Example. Klasaur 2/4**

## 324. JENSEN'S INEQUALITY I

**Jensen's inequality** is known from analysis. For a convex function  $f(x)$  it holds that

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i).$$

This inequality can also be extended to expectations.

## 325. JENSEN'S INEQUALITY II

**Theorem 9.25**

Let  $f : I \rightarrow \mathbb{R}$  be a convex function on an interval  $I \subset \mathbb{R}$  and  $X : \Omega \rightarrow I$  an integrable random variable. Then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Let  $g : I \rightarrow \mathbb{R}$  be a concave function on an interval  $I \subset \mathbb{R}$  and  $X : \Omega \rightarrow I$  an integrable random variable. Then

$$g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)].$$

## 327. JENSEN'S INEQUALITY IV

**Example 9.26**

- (1)  $\mathbb{E}(X^2) \geq \mathbb{E}(X)^2$ , since  $f(x) = x^2$  is convex. (Hence variance is always non-negative.)  
 (2) If  $\mathbb{P}(X > 0) = 1$ , then

$$\mathbb{E}\left(\frac{1}{X}\right) \geq \frac{1}{\mathbb{E}(X)} \quad \text{with } f(x) = \frac{1}{x}.$$

**Extra Examples:****Exponential function**

Take  $f(x) = e^x$ , convex on  $\mathbb{R}$ .

Then

$$e^{\mathbb{E}[X]} \leq \mathbb{E}[e^X].$$

**Absolute value**

Let  $f(x) = |x|$ , which is convex.

Then

$$|\mathbb{E}[X]| \leq \mathbb{E}[|X|].$$

**Logarithm**

For concave  $g(x) = \ln(x)$ , Jensen's inequality reverses:

$$\ln(\mathbb{E}[X]) \geq \mathbb{E}[\ln(X)], \quad X > 0.$$

**Klasaur 1/6**

$$f(x) = \frac{3}{2}(x-1)^2 \cdot I_{[0,2]}(x)$$

the continuous density of the random variable  $X$ .

Estimate the expectation of  $Z = \sin(X)$ . (No explicit calculation!)

**Solution:**  $\sin(X)$  is concave on  $[0, 2]$ , so Jensen's inequality reverses.

$\mathbb{E}[X] = 1$  because the density is symmetric around 1.

$\mathbb{E}[Z] \leq \sin(\mathbb{E}[X]) = \sin(1) = 0.841$ .

## 329. NORMS I

**Definition 0.29** (9.27 (Norm)). Let  $f : \Omega \rightarrow \mathbb{R}$  be  $\mathcal{F}$ - $\mathcal{B}$ -measurable on  $(\Omega, \mathcal{F}, \mu)$ . Then

$$\|f\|_p := \left( \int |f|^p d\mu \right)^{1/p} \in [0, \infty[$$

is called the  $p$ -norm of  $f$ .

- For a random variable  $X$ ,  $\|X\|_p$  is the  $p$ -th root of the  $p$ -th absolute moment of  $X$ :

$$\mathbb{E}(|X|^p) = \|X\|_p^p.$$

## 342. CONTINUOUS DISTRIBUTIONS I

**Definition 0.30** (10.1). A real random variable  $X$  is called continuously distributed if there exists a nonnegative Borel-measurable function  $f : \mathbb{R} \rightarrow \mathbb{R}$  with the property

$$\int_{-\infty}^{\infty} f(t) dt = 1,$$

such that

$$\mathbb{P}_X(B) = \mathbb{P}(X \in B) = \int_B f(t) dt, \quad B \in \mathcal{B}.$$

The function  $f$  is called the continuous density (with respect to Lebesgue measure) of  $X$  or of  $\mathbb{P}_X$ .

## 343. CONTINUOUS DISTRIBUTIONS II

- $f$  does not need to be continuous, but there must be a continuous function  $g : \mathbb{R} \rightarrow \mathbb{R}$  such that  $f = g \cdot 1_{[a,b]}$ .
- $T = [a, b]$  is the support of the continuous distribution.
- The distribution function  $F_{\mathbb{P}}$  is continuous on all of  $\mathbb{R}$ .

**Definition 10.2**

Let  $\mathbb{P}$  be any distribution with density  $f(x)$ , then

$$T = \{x \mid f(x) > 0\}$$

is the support of the distribution.

This definition applies to any distribution, see Def. 6.1 for discrete distributions.

**Example (discrete):** Support of a fair die  $= \{1, 2, 3, 4, 5, 6\}$ .

**Example (continuous):** Support of uniform(0,1)  $= [0, 1]$ .

**Example (normal):** Support is all of  $\mathbb{R}$ , since  $f(x) > 0$  for all  $x$ .

## 344. REMINDER: DENSITY TRANSFORMATION THEOREM

**Theorem 10.3 (Density Transformation Theorem)**

Let  $X : \Omega \rightarrow \mathbb{R}$  be a random variable with continuous distribution function  $F_X$  and density

$$f_X(x) = \frac{\partial F_X(x)}{\partial x}$$

with respect to the Lebesgue measure  $\lambda$ . Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be bijective and continuously differentiable with  $\frac{dg(x)}{dx} \neq 0$ , and let  $h = g^{-1}$ . Then  $g \circ X$  has the density

$$f_{g \circ X}(y) = f_X(h(y)) \cdot \left| \frac{\partial h(y)}{\partial y} \right|$$

with respect to  $\lambda$ .

- Generalization with proof later.

**Example 1: Linear transformation**

Suppose  $X \sim \text{Uniform}(0, 1)$  and let  $Y = g(X) = 2X + 3$ . Here:

- $g(x) = 2x + 3$ ,
- inverse  $h(y) = \frac{y-3}{2}$ ,
- derivative  $h'(y) = \frac{1}{2}$ .

The density of  $X$  is

$$f_X(x) = 1 \cdot 1_{[0,1]}(x).$$

So by the theorem:

$$f_Y(y) = f_X\left(\frac{y-3}{2}\right) \cdot \left|\frac{1}{2}\right| = \frac{1}{2} \cdot 1_{[3,5]}(y).$$

Thus  $Y \sim \text{Uniform}(3, 5)$ .

**Example 2: Exponential distribution**

Suppose  $X \sim \text{Exp}(1)$  with density  $f_X(x) = e^{-x}1_{(0,\infty)}(x)$ .

Let  $Y = g(X) = e^X$ . Then:

- inverse:  $h(y) = \ln(y)$ ,
- derivative:  $h'(y) = \frac{1}{y}$ .

So:

$$f_Y(y) = f_X(\ln(y)) \cdot \frac{1}{y} = e^{-\ln(y)} \cdot \frac{1}{y}, \quad y > 1.$$

Simplify:

$$f_Y(y) = \frac{1}{y^2}, \quad y > 1.$$

## 362. EXPONENTIAL DISTRIBUTION IV

**Theorem 10.11 (Memorylessness of the Exponential Distribution)**

Let  $X$  be an exponential random variable with parameter  $\lambda$ . Then for all  $s, t \geq 0$ ,

$$\mathbb{P}(X > t + s \mid X > t) = \mathbb{P}(X > s).$$

This means that the probability of the event happening in the next  $s$  time units is independent of the time that has already passed.

## 357. EXPONENTIAL DISTRIBUTION I

*Exponential distribution*

If  $X \sim \text{Exp}(\lambda)$  with  $\lambda > 0$ , then the density is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

It models **waiting times** until an event occurs, under the assumption that events happen randomly and independently at a constant rate.

- *Waiting times*
  - Time until the next bus arrives (if arrivals follow a Poisson process).
  - Time until a light bulb burns out (assuming constant failure rate).
  - Time between radioactive particle emissions.
- *Lifetimes*
  - Lifespan of a component with no “aging” (the chance of failure doesn’t increase or decrease with age).
  - Survival time of a system with a constant hazard rate.

- *Connection with Poisson process:* If arrivals follow a Poisson process with rate  $\lambda$ , then the interarrival times (the gaps between events) are  $\text{Exp}(\lambda)$ -distributed.
- $\lambda$  represents the rate at which events occur per unit time. Example: if  $\lambda = 2$ , then on average 2 events happen per unit of time.
- *Expected waiting time:* The mean of the exponential distribution is

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

- Small  $\lambda$  means rare events, long waiting times.
- Large  $\lambda$  means frequent events, short waiting times.

### 363. EXPONENTIAL DISTRIBUTION V

- The exponential distribution is the simplest assumption for waiting times.
- The "memorylessness" property implies that the future waiting time is independent of how much time has already elapsed.

### 364. LIFETIMES I

#### **Definition 10.12**

Let  $T$  be a continuous non-negative random variable (waiting time for an event). Then:

$$S(t) := \mathbb{P}(T \geq t) = 1 - F_T(t)$$

is the **survivor function**, and

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t \leq T \leq t + \Delta t \mid t \leq T)}{\Delta t}$$

is the **hazard rate** (sometimes denoted by  $\lambda(t)$ ).

### 365. LIFETIMES II

- The survivor function is the probability to "survive" until time  $t$ .
- The hazard rate is the probability that the event happens "at the next moment."
- It holds:

$$h(t) = \frac{f(t)}{S(t)}$$

- If the waiting time  $T \sim \text{Exp}(\lambda)$  is exponentially distributed, then the hazard rate is constant. It holds for  $t > 0$ :

$$h(t) = \frac{\lambda \exp(-\lambda t)}{\exp(-\lambda t)} = \lambda$$

## 366. LIFETIMES III

**Example 10.13** The bus arrives randomly, on average every ten minutes. Let  $X$  be the waiting time for the bus in minutes. (*Incorrect in the lecture notes!*)

- $X \sim \text{Exp}\left(\frac{1}{10}\right)$
- $h(x) = \frac{1}{10}$

**Example 10.14** A bus arrives (punctually) every 10 minutes. You arrive at a random time at the bus stop ( $Y \sim \mathcal{U}(0, 10)$ ).

- $X = (10 - Y) \sim \mathcal{U}(0, 10)$
  - $f(x) = 0.1$  for  $x \in [0, 10]$
  - $S(x) = 1 - 0.1x = 0.1 \cdot (10 - x)$  for  $x \in [0, 10]$
  - $h(x) = \frac{f(x)}{S(x)} = \frac{1}{10-x}$  for  $x \in [0, 10]$
- 

## 373. GAMMA DISTRIBUTION I

See Statistics 1 and Def. 9.23.

**Definition 0.31** (10.17). Let  $X : \Omega \rightarrow \mathbb{R}$  be a random variable with continuous density

$$f_X(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx) \cdot I_{(0,\infty)}(x)$$

then  $X$  is called Gamma distributed:  $X \sim \text{Ga}(a, b)$  with parameters  $a, b > 0$ .

- In the Anglo-Saxon world, the parameterization with scale parameter  $\theta$  is common:

$$f_X(x) = \frac{1}{\theta^a \Gamma(a)} x^{a-1} \exp(-x/\theta) \cdot I_{(0,\infty)}(x).$$


---

## 380. POISSON PROCESS I

The exponential distribution is the simplest model for waiting times. For example, suppose in a machine a component has a lifetime  $X_i \sim \text{Exp}(\lambda)$ . When the component fails, it is replaced and the waiting time starts anew. The  $X_i$  are stochastically independent. How many components are needed in a time interval  $[0, t]$ ?

$$S_n := \sum_{i=1}^n X_i$$

is the time until the  $n$ -th event. According to Theorem 10.18 and Theorem 10.19, we have

$$S_n \sim \text{Ga}(n, \lambda).$$

Define

$$N_t := \sup\{k \in \mathbb{N}_0 \mid S_k \leq t\}$$

to be the number of failures up to time  $t$ . What is the distribution of  $N_t$ ?

---

## 382. POISSON PROCESS III

Thus,  $N_t \sim \text{Po}(\lambda)$  for all  $t \in \mathbb{R}_0^+$ ! This is not just a single random variable, but an entire family  $(N_t)_{t \geq 0}$  of (uncountably many) random variables. This is called a **stochastic process**, in this case the **Poisson process**.

More about stochastic processes later.

---



## 418. UNIQUENESS

**Theorem 11.7**

Let  $\varphi_X$  and  $\varphi_Y$  be characteristic functions. Then:

$$\varphi_X = \varphi_Y \iff \mathbb{P}_X = \mathbb{P}_Y.$$

Proof see 16.4.7 in @Schmidt2009.

**Theorem 11.8**

Let  $X$  and  $Y$  be two random variables with moment-generating functions  $M_X$  and  $M_Y$ , where  $M_X(s), M_Y(s) < \infty$  for all  $s \in (-a, a)$ . Then  $\mathbb{P}_X = \mathbb{P}_Y$  if and only if  $M_X(s) = M_Y(s)$  for all  $s \in (-a, a)$ .

## 420. INDEPENDENCE

**Theorem 11.10**

Let  $X_1, \dots, X_n$  be independent real random variables with moment-generating functions  $M_1, \dots, M_n$  defined on the interval  $] -a, a[$ , and let  $\varphi_{X_1}, \dots, \varphi_{X_n}$  be their characteristic functions. Then, for  $S := \sum_{i=1}^n X_i$ , the moment-generating function  $M(s)$  is defined on  $] -a, a[$  by

$$M(s) = \prod_{i=1}^n M_i(s), \quad s \in ] -a, a[.$$

Moreover, the characteristic function of  $S$  is

$$\varphi_S = \prod_{i=1}^n \varphi_i.$$

Proof later.

## 428. GOAL

We will now consider sequences of random variables (measurable mappings)

$$X_1, X_2, \dots, X_n : \Omega \rightarrow \mathbb{R}$$

- What happens as  $n \rightarrow \infty$ ?
- What does convergence mean for random variables or distributions?
- What types of convergence theorems exist for random variables, and what are the implications?

## 430. LAW OF LARGE NUMBERS SIMULATION

**Law of Large Numbers Simulation.** The R code

$$x \leftarrow \text{rnorm}(1000, 0, 1), \quad M_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad n = 1, \dots, 1000$$

generates 1000 i.i.d. standard normal random variables  $X_i \sim \mathcal{N}(0, 1)$  and plots the sequence of running sample means  $(M_n)_{n=1}^{1000}$ .

The cumulative sum `cumsum(x)` produces the partial sums  $S_n = X_1 + \dots + X_n$ , and dividing by  $n$  gives  $M_n = S_n/n$ . The resulting plot shows the fluctuations of  $M_n$  as  $n$  increases.

Since  $\mathbb{E}[X_i] = 0$  and  $\text{Var}(X_i) = 1 < \infty$ , the Strong Law of Large Numbers applies, yielding

$$M_n \xrightarrow{\text{a.s.}} 0 \quad \text{as } n \rightarrow \infty.$$

Thus the running average stabilizes near the true mean 0, illustrating the Law of Large Numbers.

## 431. EMPIRICAL DISTRIBUTION FUNCTION (ECDF).

Given a sample  $x_1, \dots, x_n$ , the empirical distribution function is defined as

$$F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{x_i \leq t\}}, \quad t \in \mathbb{R}.$$

Thus  $F_n(t)$  is the fraction of observations less than or equal to  $t$ . The function  $F_n$  is a step function that jumps by  $1/n$  at each data point.

- `pnorm(...)` computes the theoretical normal CDF  $\Phi(t)$ .
- `lines(...)` overlays the true CDF (dashed blue line) for comparison.

**Glivenko–Cantelli Theorem.**

For i.i.d. samples from distribution function  $F$ , we have

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \xrightarrow{a.s.} 0 \quad \text{as } n \rightarrow \infty.$$

Hence the empirical distribution converges uniformly to the true distribution function.

## 432. COMPARISON OF EMPIRICAL AND THEORETICAL CDFs.

**Comparison of Empirical and Theoretical CDFs.**

In the R code

```
plot(ecdf(x), main = "Empirische Verteilungsfunktion")
lines(seq(-5, 5, length = 1000),
      pt(seq(-5, 5, length = 1000), 2),
      col = "blue", lty = 2)
```

the command `ecdf(x)` computes the empirical distribution function

$$F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{x_i \leq t\}},$$

based on the sample  $x_1, \dots, x_n$ . This is plotted as a step function (black line).

The command `pt(..., 2)` evaluates the cumulative distribution function of the Student  $t$ -distribution with 2 degrees of freedom:

$$F_{t_2}(t) = \mathbb{P}(T \leq t), \quad T \sim t_2.$$

The result is added as a smooth dashed blue line for comparison.

**Why does it match better?**

The ECDF must converge to the normal CDF (not  $t_2$ ) as  $n \rightarrow \infty$ . With  $n = 1000$ , the sample variability in the tails can make the ECDF by chance look closer to a heavier-tailed distribution. That's why you're seeing this effect, but it will disappear with larger  $n$ .

## 436. TYPES OF CONVERGENCE I

From analysis we know pointwise convergence

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad \forall x$$

and uniform convergence

$$\lim_{n \rightarrow \infty} \sup_x |f_n(x) - f(x)| = 0.$$

or equivalently

$$\forall \varepsilon > 0 \exists N \text{ such that } \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$$

- *Theorem: Uniform convergence implies pointwise convergence, but not vice versa.*

#### 437. TYPES OF CONVERGENCE II

##### **Example 12.1**

$$f_n : [0, 1] \rightarrow \mathbb{R}, \quad f_n(x) = x^n$$

$$f(x) = \begin{cases} 0 & \text{for } 0 \leq x < 1, \\ 1 & \text{for } x = 1 \end{cases}$$

It holds:  $f_n(x)$  converges pointwise to  $f(x)$ , but not uniformly.

For random variables, even pointwise convergence can sometimes be too strong.

- We define four types of convergence for random variables:
  - Almost sure convergence
  - Convergence in expectation
  - Convergence in probability
  - Convergence in distribution (in the chapter after next)

#### 438. ALMOST SURE CONVERGENCE I

##### **Idea:**

- What we actually want is pointwise convergence of  $X_n(\omega) \rightarrow X(\omega)$ .
- But “impossible” outcomes do not concern us.
- So let the event  $A$  be: “The set where  $X_n$  does not converge to  $X$ ” is a  $\mathbb{P}$ -null set.
- The statement “ $X_n$  converges pointwise to  $X$ ” therefore holds almost everywhere.

**Definition 0.32** (12.2). Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and  $X_i : \Omega \rightarrow \mathbb{R}$  random variables,  $i = 1, 2, \dots$ . Then the sequence  $(X_n)_{n \in \mathbb{N}}$  converges almost surely to  $X : \Omega \rightarrow \mathbb{R}$  if

$$\mathbb{P}(\{\omega \in \Omega \mid X_n(\omega) \rightarrow X(\omega) \text{ as } n \rightarrow \infty\}) = 1.$$

We write  $X_n \xrightarrow{a.s.} X$  or  $X_n \rightarrow X$  with probability 1.

#### 439. ALMOST SURE CONVERGENCE II

##### **Example 12.3**

Let  $X_1 \sim U[0, 1]$  and  $X_n := X_1^n$ . Then  $X_n \xrightarrow{a.s.} 0$ .

**Proof:** Without loss of generality, let

$$([0, 1], \mathcal{B}_{[0,1]}, U[0, 1])$$

and  $X_1(\omega) = \omega$ . Then for every  $0 \leq \omega < 1$ :

$$\lim_{n \rightarrow \infty} \omega^n = 0.$$

For  $\omega = 1$  we have  $\omega^n = 1^n = 1$ .

Thus,

$$A := \{\omega \in \Omega \mid X(\omega)^n \not\rightarrow 0\} = \{1\},$$

which is a null set with respect to the continuous uniform distribution.

- In this example, the sequence of random variables converges to a scalar (which is usually the case in practice).
- We can also view this as convergence to a random variable with a Dirac distribution at this scalar.

#### 440. CONVERGENCE IN PROBABILITY I

**Idea:**

In uniform convergence, we consider the distance of the sequence elements from the function to which the sequence converges:

$$\forall \varepsilon > 0 \exists N, \text{ such that } \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$$

- For random variables, it is sufficient that this distance is less than  $\varepsilon$  with probability 1.
- Here we consider the event

$$A_n = \{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\},$$

i.e., that  $X_n$  deviates from  $X$  by more than  $\varepsilon$ .

- For arbitrarily small  $\varepsilon > 0$ , we require that  $\mathbb{P}(A_n)$  converges to zero.

#### 441. CONVERGENCE IN PROBABILITY II

**Definition 0.33** (12.4). Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and  $X_i : \Omega \rightarrow \mathbb{R}$ ,  $i = 1, 2, \dots$ . Then the sequence  $(X_n)_{n \in \mathbb{N}}$  converges in probability to  $X : \Omega \rightarrow \mathbb{R}$  if

$$\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad \forall \varepsilon > 0.$$

We write  $X_n \xrightarrow{\mathbb{P}} X$  and also speak of stochastic convergence.

**Example 12.5**

As above, let  $X_1 \sim U[0, 1]$  and  $X_n := X_1^n$ . Then  $X_n \xrightarrow{\mathbb{P}} 0$ .

Without loss of generality, let  $([0, 1], \mathcal{B}_{[0,1]}, U[0, 1])$  and  $X(\omega) = \omega$ . Then

$$\mathbb{P}(|X_n - 0| > \varepsilon) = \mathbb{P}(X_1^n > \varepsilon) = \mathbb{P}(X_1 > \varepsilon^{1/n}) = 1 - \varepsilon^{1/n} \xrightarrow[n \rightarrow \infty]{} 0.$$

#### 442. CONVERGENCE IN PROBABILITY III

It is no coincidence that both types of convergence apply here:

**Theorem 0.17** (12.6). Let  $(X_n)_{n \in \mathbb{N}}$  be a sequence of random variables  $X_i : \Omega \rightarrow \mathbb{R}$  and  $X : \Omega \rightarrow \mathbb{R}$ . Then

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{\mathbb{P}} X$$

#### 443. CONVERGENCE IN PROBABILITY III

- The converse does not hold!
- Almost sure convergence is therefore **stronger** than convergence in probability.

## 444. CONVERGENCE IN PROBABILITY V

**Example 12.7**

Probability space  $([0, 1], \mathcal{B}_{[0,1]}, U[0, 1])$ ,  $X(\omega) = \omega$ . Let  $m \geq 0$ ,  $0 \leq k \leq 2^m - 1$  and  $n = 2^m + k$ . Further, let

$$A_n = [k2^{-m}, (k+1)2^{-m}) \quad \text{and} \quad X_n(\omega) = 1_{A_n}(\omega).$$

Then for  $0 < \varepsilon < 1$ :

$$\mathbb{P}(|X_n| > \varepsilon) = 2^{-m}.$$

For  $n \rightarrow \infty \Rightarrow m \rightarrow \infty \Rightarrow 2^{-m} \rightarrow 0$ , hence  $X_n \xrightarrow{\mathbb{P}} 0$ .

However,  $X_n$  does not converge almost surely, since for every  $\omega \in [0, 1]$  the sequence  $X_n(\omega)$  does not converge.

$$X_1(\omega) = 1,$$

$$X_2(\omega) = 1_{[0, 1/2)}(\omega),$$

$$X_3(\omega) = 1_{[1/2, 1)}(\omega),$$

$$X_4(\omega) = 1_{[0, 1/4)}(\omega),$$

$$X_5(\omega) = 1_{[1/4, 1/2)}(\omega),$$

$$X_6(\omega) = 1_{[1/2, 3/4)}(\omega),$$

$$X_7(\omega) = 1_{[3/4, 1)}(\omega).$$

445. CONVERGENCE IN THE  $r$ -TH MOMENT I**Idea:**

- We again consider the distance of the sequence elements from the random variable to which convergence takes place:  $|X_n - X|$ .
- This is itself a random variable.
- The distance does not need to converge to zero, but at least its expectation should converge to zero:

**Definition 0.34** (12.8). Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and  $X_i : \Omega \rightarrow \mathbb{R}$ ,  $i = 1, 2, \dots$ . Then the sequence  $(X_n)_{n \in \mathbb{N}}$  converges in the  $r$ -th moment (or in the  $r$ -th mean),  $r \geq 1$ , to  $X : \Omega \rightarrow \mathbb{R}$ , if

$$\mathbb{E}(|X_n|^r) < \infty \quad \forall n$$

and

$$\mathbb{E}(|X_n - X|^r) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

We write  $X_n \xrightarrow{r} X$  or  $X_n \xrightarrow{L^r} X$ .

446. CONVERGENCE IN THE  $r$ -TH MOMENT II**Example 12.9**

Let  $r > s \geq 1$  and

$$X_n = \begin{cases} n & \text{with probability } n^{-\frac{1}{2}(r+s)}, \\ 0 & \text{with probability } 1 - n^{-\frac{1}{2}(r+s)}. \end{cases}$$

We compute:

$$\mathbb{E}(|X_n|^s) = n^s \cdot n^{-\frac{1}{2}(r+s)} = n^{\frac{1}{2}(s-r)} \longrightarrow 0,$$

since  $s < r$ . Thus  $X_n \xrightarrow{s} 0$ .

Furthermore,

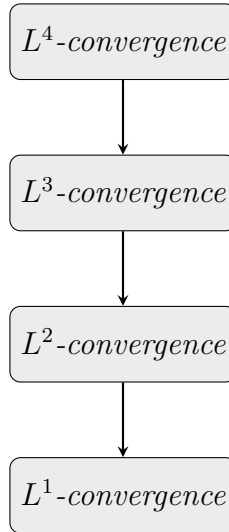
$$\mathbb{E}(|X_n|^r) = n^r \cdot n^{-\frac{1}{2}(r+s)} = n^{\frac{1}{2}(r-s)} \longrightarrow \infty,$$

since  $r > s$ .

#### 447. CONVERGENCE IN THE $r$ -TH MOMENT III

**Theorem 0.18** (12.10). Let  $(X_n)_{n \in \mathbb{N}}$  be a sequence of random variables  $X_i : \Omega \rightarrow \mathbb{R}$  and  $X : \Omega \rightarrow \mathbb{R}$ . Then the following holds:

$$\forall r > s \geq 1 : \quad X_n \xrightarrow{r} X \implies X_n \xrightarrow{s} X$$



Higher moment convergence implies lower moment convergence, but not the other way around.

#### 448. CONVERGENCE IN THE $r$ -TH MOMENT IV

Perhaps somewhat surprisingly, convergence in the  $r$ -th moment is stronger than convergence in probability:

**Theorem 0.19** (12.11). Let  $(X_n)_{n \in \mathbb{N}}$  be a sequence of random variables  $X_i : \Omega \rightarrow \mathbb{R}$  and  $X : \Omega \rightarrow \mathbb{R}$ . Then for  $r \geq 1$  it holds:

$$X_n \xrightarrow{L^r} X \implies X_n \xrightarrow{\mathbb{P}} X$$

**Proof.**

By Theorem 12.10 it suffices to take  $r = 1$ . Then

$$\mathbb{P}(|X_n - X| > \varepsilon) \leq \frac{\mathbb{E}(|X_n - X|)}{\varepsilon},$$

using Markov's inequality applied to  $(X_n - X)$ . □

#### 449. OVERVIEW OF RELATIONSHIPS

- Almost sure convergence  $\implies$  convergence in probability
- Convergence in the  $r$ -th moment  $\implies$  convergence in probability
- For  $r > s$ : Convergence in the  $r$ -th moment  $\implies$  convergence in the  $s$ -th moment

**Theorem 0.20** (12.12). *If a sequence of random variables  $X_n$  converges in the  $r$ -th mean to  $X$  and*

$$\sum_{n=1}^{\infty} \mathbb{E}(|X_n - X|^r) < \infty,$$

*then the sequence also converges almost surely to  $X$ .*

#### 450. LAW OF LARGE NUMBERS

##### 451. I.I.D. MODEL

*We consider random variables  $X_1, \dots, X_n$  that are i.i.d. (independently, identically distributed), i.e., they*

- (1) are independent and*
- (2) have the same distribution.*

*Furthermore, the expectation  $\mu$  and the variance  $\sigma^2$  exist. This particularly represents the situation in which  $X_1, \dots, X_n$  form a sample of a characteristic  $\tilde{X}$  in a simple random selection.*

**Example:**  $\tilde{X}$  is the income,  $n$  people are selected at random

$X_1$  income of the first randomly selected person

$X_2$  income of the second randomly selected person

$\vdots$

$X_n$  income of the  $n$ -th randomly selected person

---

#### 452. WEAK LAW OF LARGE NUMBERS

**Theorem 13.1** *Let  $X_1, \dots, X_n$  be independent and identically distributed random variables with  $\mathbb{E}(X_i) = \mu$  and  $\text{Var}(X_i) = \sigma^2 < \infty$  for all  $i = 1, \dots, n$ .*

*Then, it holds that:*

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu.$$


---

#### 453. WEAK LAW OF LARGE NUMBERS II

**Proof.**

$$\bar{X}_n \xrightarrow{2} \mu$$

since

$$\mathbb{E}(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = \frac{1}{n} \cdot n\mu = \mu$$

and

$$\mathbb{E}(|\bar{X}_n - \mu|^2) = \text{Var}(\bar{X}_n) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{1}{n}\sigma^2 \longrightarrow 0.$$

$$\implies \bar{X}_n \xrightarrow{\mathbb{P}} \mu \quad \text{by Theorem 12.11.}$$


---

## 454. WEAK LAW OF LARGE NUMBERS III

**Example 13.2**

$$X_i \sim \text{Po}(\lambda) \text{ i.i.d.} \implies \mathbb{E}(X_i) = \lambda, \text{ Var}(X_i) = \lambda, \quad i = 1, \dots, n$$

and

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \lambda.$$


---

## 456. BERNOULLI'S THEOREM I

*Special case of the law of large numbers: We consider  $n$  independent 0/1-experiments with success probability  $\pi$ . That is,  $X_i \sim \text{B}(1, \pi)$ . Then  $\bar{X}$  is the relative frequency of successes, and we have*

$$\bar{X}_n \xrightarrow{\mathbb{P}} \pi$$

*Thus, the relative frequency converges to the true probability.*

---

## 458. KHINCHIN'S THEOREM I

*Khinchin's Theorem (without proof): The assumption  $\text{Var}(X_i) = \sigma^2$  is not necessary! It suffices that  $\mathbb{E}(X_i)$  exists.*

*For example, let  $X_i \sim t(n)$  (Student's  $t$  distribution):*

- $\mathbb{E}(X_i) = 0$  for  $n > 1$ ; for  $n = 1$  the expectation does not exist.
  - $\text{Var}(X_i) = \frac{n}{n-2}$  for  $n > 2$ ; for  $n \leq 2$  the variance does not exist.
- 

**Theorem 0.21** (Khinchin's Weak Law of Large Numbers). *Let  $X_1, X_2, \dots$  be i.i.d. random variables with finite expectation*

$$\mathbb{E}[X_1] = \mu.$$

*Then the sample mean*

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

*converges in probability to  $\mu$ :*

$$\bar{X}_n \xrightarrow{\mathbb{P}} \mu.$$

**Remarks:**

- *No assumption on the variance is needed (contrast with Chebyshev's proof of the WLLN).*
  - *This makes the result applicable to heavy-tailed distributions such as Student's  $t$  with  $1 < \nu \leq 2$ .*
  - *Stronger versions (Kolmogorov, Marcinkiewicz-Zygmund) give almost sure convergence under additional conditions.*
-



## 461. COUNTEREXAMPLE I

**Example 13.3**

Let  $X_i \sim \text{Cauchy}$  i.i.d. The law of large numbers does not apply to

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i,$$

since the Cauchy distribution has no expectation!

## 473. CONVERGENCE IN DISTRIBUTION I

**Definition 14.1**

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $X_i : \Omega \rightarrow \mathbb{R}$  be random variables with distribution functions  $F_i$ ,  $i = 1, 2, \dots$ .

Then the sequence  $(X_n)_{n \in \mathbb{N}}$  converges in distribution to  $X : \Omega \rightarrow \mathbb{R}$  with distribution function  $F$ , if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x)$$

for all  $x$  at which  $F$  is continuous.

We write  $X_n \xrightarrow{D} X$ . This is also called weak convergence.

## 474. CONVERGENCE IN DISTRIBUTION II

**Example**

Let  $X \sim U(0, 1)$ . Define  $Z_n = 1 - X$ . Then

$$Z_n \xrightarrow{D} X.$$

Of course, we also have, for example,

$$Z_n \xrightarrow{D} Z_1.$$

We also write

$$Z_n \xrightarrow{D} U(0, 1).$$

- The convergence here is indeed towards a random variable.
- However, the random variable is not unique.
- The distribution of the random variable is unique.

## 486. CENTRAL LIMIT THEOREM I

**Theorem 14.4 (Central Limit Theorem of Lindenberg–Lévy)**

Let  $X_i$ ,  $i = 1, \dots$  be independent and identically distributed (iid) with

$$\mathbb{E}(X_i) = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty, \quad \forall i = 1, \dots$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Then we have

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} N(0, 1)$$

or equivalently

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{D} N(0, \sigma^2).$$

## 487. EXAMPLE I

**Example 14.5 (Discrete Uniform Distribution — Dice Roll)**

Let  $X_i \sim U\{1, 2, 3, 4, 5, 6\}$ . Then

$$\mathbb{E}(X_i) = 3.5 \quad \text{and} \quad \text{Var}(X_i) = \frac{35}{12}.$$

Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, by the Central Limit Theorem,

$$\sqrt{n} \frac{\bar{X} - 3.5}{\sqrt{35/12}} \xrightarrow{D} N(0, 1).$$

Central Limit Theorem Example: Klasaur 1/7, 2/4, 3/8

Normal Distribution Example: Klasaur 2/9e

Density Transformation Example: Klasaur 3/5e

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505.  $n$ -FOLD COIN TOSS**Example 2.2**

We toss the coin  $n$  times. Outcomes are  $n$ -tuples of the form

$$\omega = (1, 0, 0, \dots, 1).$$

The sample space  $\Omega_n$  consists of  $2^n$  elements.

We are interested in the probability of events  $A \subset \Omega$ .

$$\implies \quad \text{i.i.d. model!}$$


---

## 506. SURVEY

- We ask  $n$  people: “Which party do you want to vote for?”
  - We ask one person about party preference, level of education, income, gender, and shoe size
- 

## 512. CARTESIAN PRODUCT III

**Generalization**

- The Cartesian product of the sets  $\Omega_1, \Omega_2, \dots, \Omega_n$  is denoted by

$$\prod_{i=1}^n \Omega_i = \Omega_1 \times \Omega_2 \times \dots \times \Omega_n$$

and consists of all possible  $n$ -tuples that can be formed (taking order into account) from elements of  $\Omega_1, \Omega_2, \dots, \Omega_n$ . The sets  $\Omega_1, \Omega_2, \dots, \Omega_n$  do not have to be finite; for finite sets it holds that

$$\left| \prod_{i=1}^n \Omega_i \right| = |\Omega_1| \cdot |\Omega_2| \cdot \dots \cdot |\Omega_n|.$$

- Cartesian products are used to combine results of complex experiments from individual experiments.
- Example:  $n$ -fold coin toss:  $\Omega = (\omega_1, \dots, \omega_n)$  with  $\omega_i \in \{\text{Heads}, \text{Tails}\}$ .

513. PRODUCT  $\sigma$ -ALGEBRA I

If  $\Omega = \prod_{i=1}^n \Omega_i$  and  $\mathcal{F}_i$  is a  $\sigma$ -algebra on  $\Omega_i$  for all  $i$ , then a  $\sigma$ -algebra for  $\Omega$  can be defined as follows:

$$\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes \dots \otimes \mathcal{F}_n := \sigma \left( \left\{ \prod_{i=1}^n A_i \mid A_i \in \mathcal{F}_i, i = 1, \dots, n \right\} \right).$$

514. PRODUCT- $\sigma$ -ALGEBRA II**Example 15.1**

We roll a die onto a line of length 1. The outcomes are the face values  $\Omega_1 = \{1, \dots, 6\}$  and where on the line the die lands  $\Omega_2 = (0, 1)$ .

So altogether

$$\Omega = \{(\omega_1, \omega_2) \mid \omega_1 \in \Omega_1, \omega_2 \in \Omega_2\}.$$

The generator of the corresponding  $\sigma$ -algebra is

$$\{A_1 \times A_2 \mid A_1 \in \mathcal{P}(\Omega_1), A_2 \in \mathcal{B}_{(0,1)}\}.$$

## 524. FUBINI'S THEOREM II

Using Fubini's Theorem, one defines multidimensional densities with respect to the dominating product measure  $\mu \otimes \nu$ :

- Example "Die on the Line": dominating measure is  $\mu_Z \otimes \lambda$
- For multidimensional continuous distributions,  $\lambda^n$  is the dominating measure
- For multidimensional discrete distributions,  $\mu_Z$  is the dominating measure

**Example 15.9** Uniform distribution on  $(0, 1) \times (0, 1)$ . The density is  $f(x_1, x_2) = I_{(0,1) \times (0,1)}(x_1, x_2)$ .

## 525. MULTINOMIAL DISTRIBUTION I

Urn model with  $k$  colors.  $n$  balls are drawn with replacement. The random variable  $X = (X_1, X_2, \dots, X_k)$  describes the drawn colors, where  $X_i$  is the number of balls of color  $i$ .

**Definition 15.10** Let  $k \leq n \in \mathbb{N}_0$  and  $p_1, \dots, p_k \in [0, 1]$  with  $\sum_{i=1}^n p_i = 1$ . The distribution with the count density

$$f(n_1, \dots, n_k) = \binom{n}{n_1, \dots, n_k} p_1^{n_1} \dots p_k^{n_k}$$

for  $\sum_{i=1}^k n_k = n$  (otherwise the count density is zero) is called the multinomial distribution  $M(n, (p_1, \dots, p_k))$  or  $M(n, \mathbf{p})$ .

$$\binom{n}{n_1, \dots, n_k} = \frac{n!}{n_1! \dots n_k!}$$

is the multinomial coefficient.

## 526. MULTINOMIAL DISTRIBUTION II

If we combine (w.l.o.g.) the colors 2 to  $k$ , the multinomial model obviously becomes a binomial model: Either color 1 is drawn with probability  $p_1$ , or it is not (with probability  $1-p_1 = \sum_{i=2}^k p_i$ ).

---

## 527. DIRICHLET DISTRIBUTION I

**Definition 15.11**

Let  $m \geq 2$ . With  $x \in [0, 1]$  and  $\sum_{i=1}^m x_i = 1$ , the distribution with the density

$$f(x|\alpha) = \frac{\Gamma(\sum_{i=1}^m \alpha_i)}{\prod_{i=1}^m \Gamma(\alpha_i)} x_1^{\alpha_1-1} x_2^{\alpha_2-1} \cdots x_m^{\alpha_m-1}$$

with respect to the  $(m-1)$ -dimensional Lebesgue measure is called the Dirichlet distribution. The support is the  $(m-1)$ -simplex.

$$\mathbf{X} \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_m).$$

A special case of the Dirichlet distribution for  $m = 2$  is the Beta distribution.

---

## 530. MARGINAL DISTRIBUTION I

**Theorem 15.12**

Let  $\mathbf{X} : \Omega \rightarrow \mathbb{R}^n$  be an  $n$ -dimensional random variable and  $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^k$  be  $\mathcal{B}^n$ - $\mathcal{B}^k$  measurable. Then

$$\mathbf{g} \circ \mathbf{X} : \Omega \rightarrow \mathbb{R}^k, \quad \omega \mapsto \mathbf{g}(\mathbf{X}(\omega))$$

is a  $k$ -dimensional real random variable with the distribution

$$\mathbb{P}_{\mathbf{g}(\mathbf{X})}(A) = \mathbb{P}(\mathbf{g}(\mathbf{X}) \in A) = \mathbb{P}(\{\omega \in \Omega | \mathbf{g}(\mathbf{X}(\omega)) \in A\}).$$

**Proof.**

The proof follows from Theorem 4.18 (Measurability of the composition). □

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## 531. MARGINAL DISTRIBUTION II

This means, in particular, that for an  $n$ -dimensional random variable or random vector  $\mathbf{X} = (X_1, \dots, X_n)$ , each  $X_i$  is also a one-dimensional random variable.

**Definition 15.13 (Marginal Distribution)**

Let  $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}$  be given by  $\mathbf{g}(\mathbf{x}) = x_j$  for a fixed  $j \in \{1, \dots, n\}$  (projection onto the  $j$ -th coordinate). Then:

$$\mathbf{g}(\mathbf{X}) = X_j,$$

$$\mathbb{P}_{X_j}(A) = \mathbb{P}_{\mathbf{X}}(\{x \in \mathbb{R}^n : x_j \in A\}) = \mathbb{P}(X_j \in A) \quad \text{for } A \in \mathcal{B}.$$

$\mathbb{P}_{X_j}$  is called the marginal distribution of  $X_j$ .

---

## 532. MARGINAL DISTRIBUTION III

- If  $X$  has the density  $f$ , then  $X_j$  (with  $\nu = \lambda$  or  $\nu = \mu_Z$ ) has the **marginal density** (or marginal distribution)

$$f_j : \mathbb{R} \rightarrow \mathbb{R}$$

$$f_j(y) = \int \cdots \int f(x_1, x_2, \dots, x_{j-1}, y, x_{j+1}, \dots, x_n) d\nu(x_n) \dots d\nu(x_{j+1}) d\nu(x_{j-1}) \dots d\nu(x_1)$$

- The expectation and variance of the marginal distribution are called the **marginal expectation** and **marginal variance** (etc.).

## 534. MARGINAL DISTRIBUTION V

**Example 15.15 (Two-Dimensional Uniform Distribution)**

$$(X, Y) \sim U(0, 1) \times U(0, 1)$$

$$f(x, y) = I_{(0,1) \times (0,1)}(x, y) = I_{(0,1)}(x) \cdot I_{(0,1)}(y)$$

$$f_X(x) = \int f(x, y) dy = I_{(0,1)}(x)$$

$$X \sim U(0, 1)$$

By symmetry:

$$Y \sim U(0, 1).$$

**Example:** Klasaur-2/2

## 535. BIVARIATE NORMAL DISTRIBUTION I

**Definition 15.16**

Let  $\mathbf{X}$  be a random variable  $\Omega \rightarrow \mathbb{R}^2$  with the density function

$$f_{\mathbf{X}}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \cdot \exp\left(-\frac{1}{2(1-\rho^2)} \left[ \frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - 2\rho \frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} \right]\right).$$

With respect to  $\lambda^2$ ,  $\mathbf{X}$  is then bivariate normally distributed:

$$\mathbf{X} \sim \mathcal{N}_2 \left( \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right).$$

- The parameter  $\rho$  is the correlation coefficient, measuring the linear dependence between  $X_1$  and  $X_2$ .
- The matrix

$$\Sigma = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{pmatrix} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

is the covariance matrix. Recall from [Section](#) that

$$\text{Cov}(X_1, X_2) = \rho\sigma_1\sigma_2$$

- *Diagonal entries: variances of  $X_1, X_2$ ; off-diagonal entries: covariance  $\text{Cov}(X_1, X_2) = \rho\sigma_1\sigma_2$ .*
- *$\rho$  determines the orientation of the elliptical contours of the bivariate normal distribution.*

**Examples where bivariate normal distributions appear:**

- **Height and Weight:** In a population,  $(X_1, X_2) = (\text{height}, \text{weight})$ . Both are approximately normal and positively correlated: taller people tend to weigh more.
- **Exam Scores:**  $(X_1, X_2) = (\text{score in mathematics}, \text{score in physics})$ . Students who do well in one subject are likely to do well in the other, leading to a positive correlation.
- **Finance:**  $(X_1, X_2) = (\text{daily return of stock A}, \text{daily return of stock B})$ . Stock returns are often modeled jointly as (multivariate) normal; correlation  $\rho$  measures co-movement.
- **Measurement Errors:** Suppose two instruments measure the same physical quantity. Their errors  $(X_1, X_2)$  may be modeled as bivariate normal with some correlation.
- **Meteorology:**  $(X_1, X_2) = (\text{temperature}, \text{humidity})$  at a location on a given day. These variables can have approximately normal joint behavior with some correlation.

**Example (Height–Weight):** Let  $(X_1, X_2) = (\text{height in cm}, \text{weight in kg})$  be bivariate normal with

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \begin{pmatrix} 170 \\ 70 \end{pmatrix}, \quad (\sigma_1, \sigma_2) = (7, 12), \quad \rho = 0.6,$$

so that

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} = \begin{pmatrix} 49 & 50.4 \\ 50.4 & 144 \end{pmatrix}, \quad \text{Cov}(X_1, X_2) = \rho\sigma_1\sigma_2 = 50.4.$$

**3-Dimensional Case:**

$$\Sigma = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \text{Cov}(X_1, X_3) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \text{Cov}(X_2, X_3) \\ \text{Cov}(X_3, X_1) & \text{Cov}(X_3, X_2) & \text{Var}(X_3) \end{pmatrix}.$$

**Example:** Klasaur-1/9

## 539. BIVARIATE NORMAL DISTRIBUTION VII

For the general bivariate normal distribution:

$$\mathbf{X} \sim \mathcal{N}_2 \left( \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right),$$

it holds that:

- $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$
- $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$

## 543. DISTRIBUTION FUNCTION I

**Theorem 15.17 (Independence of  $n$ -dimensional random variables)**

Let  $\mathbf{X} = (X_1, \dots, X_n)$  be a real  $n$ -dimensional random variable. Then,  $X_1, \dots, X_n$  are stochastically independent if and only if for all  $\mathbf{c} \in \mathbb{R}^n$ :

$$\begin{aligned}
\mathbb{P}(\mathbf{X} \leq \mathbf{c}) &= \mathbb{P}(X_1 \leq c_1, \dots, X_n \leq c_n) \\
&= \prod_{i=1}^n \mathbb{P}(X_i \leq c_i), \\
F_{\mathbf{X}}(\mathbf{c}) &= \prod_{i=1}^n F_{X_i}(c_i).
\end{aligned}$$

Suppose  $X$  and  $Y$  are independent, then

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y) = F_X(x)F_Y(y)$$

For the discrete case,

$$f_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) = f_X(x)f_Y(y)$$

For the continuous case, by definition,

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B), \quad \forall A, B \subseteq \mathbb{R} \text{ measurable.}$$

$$\mathbb{P}(X \in A, Y \in B) = \iint_{A \times B} f_{X,Y}(x, y) dx dy.$$

$$\mathbb{P}(X \in A) = \int_A f_X(x) dx, \quad \mathbb{P}(Y \in B) = \int_B f_Y(y) dy.$$

$$\iint_{A \times B} f_{X,Y}(x, y) dx dy = \int_A f_X(x) dx \int_B f_Y(y) dy = \iint_{A \times B} f_X(x)f_Y(y) dx dy.$$

Thus

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \quad \text{a.e. } x, y \in \mathbb{R}.$$

**Example 0.14.** Let  $X, Y \sim \text{Bernoulli}(1/2)$ , independent. The sum  $Z = X + Y$  takes values in  $\{0, 1, 2\}$  with

$$\mathbb{P}(Z = 0) = 1/4, \quad \mathbb{P}(Z = 1) = 1/2, \quad \mathbb{P}(Z = 2) = 1/4.$$

So,

$$Z \sim \text{Binomial}(2, 1/2).$$

This is exactly the  $\text{Binomial}(2, 1/2)$  distribution.

**Example 0.15.** Let  $X, Y \sim U(0, 1)$ , independent. Then the sum  $Z = X + Y$  has pdf

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx.$$

Here  $f_X = f_Y = 1$  on  $(0, 1)$ . So

$$f_Z(z) = \begin{cases} z, & 0 < z < 1, \\ 2 - z, & 1 \leq z < 2, \\ 0, & \text{otherwise.} \end{cases}$$

This is the triangular distribution.

Two dimensional case: **Klasaur-1/8, 2/2, 3/6**

Law of total expectation: **Klasaur-2/3, 3/7**

Others to do: **Klasaur-2/7, 2/9, 3/5e**

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## 545. DENSITY OF INDEPENDENT RANDOM VARIABLES I

**Theorem 15.18**

Let  $X_1, \dots, X_n$  be real **independent** random variables with densities  $f_{X_1}, \dots, f_{X_n}$  with respect to  $\mu = \bigotimes_{i=1}^n \mu_i$ .

Then, the joint density of  $\mathbf{X} = (X_1, \dots, X_n)^\top$  is the product of the individual densities, i.e.,

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n f_{X_i}(x_i)$$

is the density with respect to  $\mu$ .

## 555. REMINDER: TRANSFORMATION THEOREM FOR DENSITIES III

Perhaps simpler:

$$F_X(x) = \mathbb{P}(X \leq x) = 1 - \exp(-\lambda x)$$

$$\begin{aligned} F_{X^2}(y) &= \mathbb{P}(X^2 \leq y) = \mathbb{P}(X \leq \sqrt{y}) \\ &= 1 - \exp(-\lambda\sqrt{y}) \end{aligned}$$

$$f_{X^2} = \frac{d}{dy} F_{X^2} = \lambda \exp(-\lambda\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} \cdot \mathbb{I}_{[0, \infty[}(\sqrt{y})$$

**Not applicable for**

- $X \sim U_{\{-2, -1, +1, +2\}} \implies X^2 \sim U_{\{1, 4\}}$
- $X \sim N(0, 1) \implies X^2 \sim \chi^2$  — but  $g(x) = x^2$  is not bijective!
- $X \sim N_k(\mu, \Sigma) \implies g(X) \sim ??$

## 556. GENERAL TRANSFORMATION THEOREM I

**Theorem 15.23 (General Transformation Theorem)**

Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random variable with density  $f_X$  with respect to  $\lambda^n$ . Let  $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be a measurable function. Furthermore, let  $G_m \in \mathcal{B}^n, m \in M$  be a partition of  $\mathbb{R}^n$  such that

$$\mathbb{P}\left(X \in \bigcup_{m \in M} G_m\right) = 1, \quad \text{so that}$$

$$g_m := g|_{G_m} \quad \text{is bijective and continuously differentiable.}$$

Let

$$J_m(y) = \left( \frac{\partial h_{im}(y)}{\partial y_j} \right)_{i=1, \dots, n}^{j=1, \dots, n}$$

be the Jacobian matrix for  $h_m = (h_{1m}, \dots, h_{nm}) = g_m^{-1}$ .

## 557. GENERAL TRANSFORMATION THEOREM II

Then we have

$$f_{g \circ X}(y) = \sum_{m \in M} \nu_m(y)$$

with

$$\nu_m(y) = \begin{cases} f_X(h_m(y)) \cdot |\det(J_m(y))|, & \text{if } h_m(y) \in G_m, \\ 0, & \text{otherwise.} \end{cases}$$



**Reminder: Jacobian Matrix.**

$$J_f(x) := \frac{\partial(f_1, \dots, f_n)}{\partial(x_1, \dots, x_n)}(x) := \begin{pmatrix} \partial_1 f_1(x) & \cdots & \partial_n f_1(x) \\ \vdots & \ddots & \vdots \\ \partial_1 f_n(x) & \cdots & \partial_n f_n(x) \end{pmatrix}.$$


---

**558. EXAMPLE OF MULTIDIMENSIONAL TRANSFORMATION**

Let  $X_1 \sim N(0, 1)$  and  $X_2 \sim N(0, 1)$  be stochastically independent. What is the distribution of  $\mathbf{Y} = \left(\frac{X_2}{X_1}, X_1\right)$ ?

Joint density:

$$f(x_1, x_2) = \varphi(x_1)\varphi(x_2) \quad \text{due to independence.}$$

Define the transformation:

$$g(x_1, x_2) := \left(\frac{x_2}{x_1}, x_1\right)$$

The inverse transformation is:

$$h(y_1, y_2) = g^{-1}(y_1, y_2) = (y_2, y_1 y_2)$$

The Jacobian determinant is:

$$\det(J) = \det \begin{pmatrix} 0 & 1 \\ \frac{1}{y_2} & y_1 \end{pmatrix} = -y_2$$

The joint density of  $Y_1, Y_2$  is:

$$f_{Y_1, Y_2}(y_1, y_2) = \varphi(y_2)\varphi(y_1 y_2) |\det(J)|$$

Thus,

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{|y_2|}{2\pi} \exp\left(-\frac{1}{2}(y_2^2 + (y_1 y_2)^2)\right)$$

Simplifying:

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{|y_2|}{2\pi} \exp\left(-\frac{1}{2}y_2^2(1 + y_1^2)\right)$$


---

**559. TRANSFORMATION WITH DIMENSION CHANGE I**

Let  $X_1 \sim N(0, 1)$ ,  $X_2 \sim N(0, 1)$  be stochastically independent. What is the distribution of  $Z = \frac{X_2}{X_1}$ ?

- *Note:* For the general transformation theorem, we need a function  $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ , not  $\mathbb{R}^n \rightarrow \mathbb{R}^m$ !
  - *Intermediate step:* The density of  $Y = (Y_1, Y_2) = \left(\frac{X_2}{X_1}, X_1\right) = (Z, X_1)$ . Then calculate the marginal density for the distribution of  $Z$ .
-

## 568. CONVOLUTION I

**Definition 15.24**

The convolution (engl. convolution, from Lat. convolvere, to roll together)  $f * g$  of two functions  $f, g : \mathbb{R}^n \rightarrow \mathbb{C}$  is defined as

$$(f * g)(x) := \int_{\mathbb{R}^n} f(\tau)g(x - \tau) d\tau$$

Intuitively, the convolution  $f * g$  means that each value of  $f$  is replaced by the weighted average  $(f * g)(x)$ , where  $f(t)$  is weighted by  $g(x - t)$ .

**Applications**

- Moving average
- Filters for signals, images
- Density of the sum of random variables
- Kernel density estimator

## 593. CONDITIONAL PROBABILITY I

In Def. 3.24 we learned conditional probability:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

The concept can be easily extended to discrete random variables. Let  $X$  and  $Y$  be discrete random variables. Then the conditional distribution of  $X$  given  $Y = y$  is

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

## 594. CONDITIONAL PROBABILITY II

**Example 17.1**

We toss a fair coin twice. Let

- $X$  = number of coin tosses resulting in heads,
- $Y = 1$  if the first toss is heads, 0 otherwise.

$$\mathbb{P}(X = x \mid Y = 1) = \frac{\mathbb{P}(X = x, Y = 1)}{\mathbb{P}(Y = 1)} = \begin{cases} \frac{0}{0.5} = 0 & \text{for } x = 0, \\ \frac{0.25}{0.5} = 0.5 & \text{for } x = 1, \\ \frac{0.25}{0.5} = 0.5 & \text{for } x = 2. \end{cases}$$

Analogously for  $Y = 0$ .

▷ See also the memorylessness of the exponential distribution, Theorem 10.11.

## 595. CONDITIONAL DISTRIBUTION I

**Example 17.2**

We roll a die. Let  $Y$  be the random variable and  $y$  the outcome of the roll. Afterwards, we flip a fair coin  $y$  times. Let  $X$  be the number of coin tosses that show heads.

- $X \mid Y = 1 \sim B(1, \frac{1}{2})$
- $X \mid Y = 2 \sim B(2, \frac{1}{2})$
- ...

In general, we can write:

$$X \mid (Y = y) \sim B(y, \tfrac{1}{2})$$

We call this the **conditional distribution** of  $X \mid Y$ .

### 596. CONDITIONAL DISTRIBUTION II

Let us compute the joint distribution of  $(X, Y)$ . We have:

$$f_{(X,Y)}(x, y) = \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x \mid Y = y) \cdot \mathbb{P}(Y = y) = \binom{y}{x} 0.5^y \mathbf{1}_{\{0, \dots, y\}}(x) \cdot \frac{1}{6} \mathbf{1}_{\{1, \dots, 6\}}(y).$$

Thus, the marginal distribution of  $X$  is given by:

$$\mathbb{P}(X = x) = \frac{1}{6} \sum_{y=1}^6 \binom{y}{x} 0.5^y \cdot \mathbf{1}_{\{0, \dots, y\}}(x).$$

| $x$                 | 0      | 1      | 2      | 3      | 4      | 5      | 6      |
|---------------------|--------|--------|--------|--------|--------|--------|--------|
| $\mathbb{P}(X = x)$ | 0.1641 | 0.3125 | 0.2578 | 0.1667 | 0.0755 | 0.0208 | 0.0026 |

### 598. CONDITIONAL DENSITY II

#### **Example 17.3**

We roll a white billiard ball on a billiard table of length 1. The random variable  $X$  describes the point in  $[0, 1]$  where the ball comes to rest.  $X \sim U[0, 1]$

We then roll  $n$  additional red balls. Define the random variable  $Y_i$  as the indicator that the  $i$ -th red ball lies to the left of the white ball (i.e., at a point less than  $X$ ).

Obviously, we have

$$\mathbb{P}(Y_i = 1 \mid X = x) = x,$$

$$Y = \sum_{i=1}^n Y_i \sim B(n, X)$$

**But:** The conditional probability  $\mathbb{P}(Y \mid X = x)$  is not defined, since  $X$  is a continuous random variable, and therefore  $\mathbb{P}(X = x) \equiv 0$ .

### 599. CONDITIONAL DENSITY III

#### **Definition 17.4 (Conditional Density)**

Let  $(X, Y)$  be random variables with joint density  $f$ . Then the conditional density of  $X$  given  $Y$  is defined as

$$f_{X|Y}(x|y) := \begin{cases} \frac{f(x,y)}{f_Y(y)}, & \text{if } f_Y(y) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

This is the conditional density of  $X$  given  $Y$ , i.e., the density of  $X|Y$ .

- For discrete  $X$  and  $Y$ , this corresponds to the conditional probability.

## 600. CONDITIONAL DENSITY IV

**Example 17.5**

Let  $(X, Y) : \Omega \rightarrow (0, 1) \times (0, 1)$  with density (in the following always  $0 < x < 1$  and  $0 < y < 1$ )

$$f(x, y) = \frac{6}{5} (x^2 + y).$$

The marginal density of  $Y$  is

$$\begin{aligned} f_Y(y) &= \int f(x, y) dx = \int_0^1 \frac{6}{5} (x^2 + y) dx \\ &= \frac{6}{5} \int_0^1 x^2 dx + \frac{6}{5} y = \frac{6}{5} \left[ \frac{1}{3} x^3 \right]_0^1 + \frac{6}{5} y = \frac{2}{5} + \frac{6}{5} y. \end{aligned}$$


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## 601. CONDITIONAL DENSITY V

The marginal density of  $X$  is

$$f_X(x) = \int f(x, y) dy = \int_0^1 \frac{6}{5} (x^2 + y) dy = \frac{6}{5} x^2 + \frac{6}{5} \int_0^1 y dy = \frac{6}{5} x^2 + \frac{3}{5}.$$

This is indeed a density, since

$$\int f_X(x) dx = \int_0^1 \left( \frac{6}{5} x^2 + \frac{3}{5} \right) dx = \frac{6}{5} \left[ \frac{1}{3} x^3 \right]_0^1 + \frac{3}{5} = 1.$$

The conditional density of  $X|Y$  is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{\frac{6}{5}(x^2 + y)}{\frac{2}{5} + \frac{6}{5}y} = \frac{3(x^2 + y)}{1 + 3y}.$$


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## 603. BAYES' THEOREM I

**Theorem 17.6 (Bayes' Theorem for General Distributions)**

Let  $X$  and  $Y$  be random variables with densities  $f_X$  (with respect to  $\mu$ ),  $f_Y$ , and joint density  $f_{X,Y}$ . Then

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)} = \frac{f_{Y|X}(y|x)f_X(x)}{\int f_{Y|X}(y|x)f_X(x) d\mu(x)}.$$

*This is the continuous form of Bayes' theorem*

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## 604. BAYES' THEOREM II

**Proof.**

By definition of the conditional density:

$$f_{X,Y}(x, y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)$$

By definition of the marginal density:

$$f_Y(y) = \int f_{X,Y}(x, y) d\mu(x) = \int f_{Y|X}(y|x)f_X(x) d\mu(x)$$


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## 606. BAYES' THEOREM IV

**Example 17.7 (Example of Bayes)**

*Continuation of Example 17.3.*

We roll a white billiard ball on a billiard table of length 1. The random variable  $X$  describes the point on  $[0, 1]$  where the ball lands. Assumption:  $X \sim U[0, 1]$ .

We then roll  $n$  additional red balls. Let the random variable  $Y_i$  describe whether the red ball  $i$  lands to the left of the white ball (i.e., at a point less than  $X$ ). Obviously,

$$\mathbb{P}(Y_i = 1 \mid X = x) = x.$$

We know the total number  $y$  of red balls that landed to the left of  $X = x$ . With

$$Y = \sum_{i=1}^n Y_i,$$

we have

$$Y \mid X = x \sim B(n, x).$$

Can we make a statement about the distribution of  $X$  given  $Y = y$ ?

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## 607. BAYES' THEOREM V

*Bayes' theorem:*

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f_{Y|X}(y|x)f_X(x)}{\int f_{Y|X}(y|x)f_X(x) dx} \\ &= \frac{\binom{n}{y}x^y(1-x)^{n-y} \cdot \mathbf{1}_{[0,1]}(x)}{\int \binom{n}{y}x^y(1-x)^{n-y} \cdot \mathbf{1}_{[0,1]}(x) dx} \\ &= \frac{1}{B(y+1, n-y+1)} x^{(y+1)-1} (1-x)^{(n-y+1)-1} \end{aligned}$$

Thus,

$$X \mid Y = y \sim \text{Beta}(y+1, n-y+1). \quad (\text{cf. Def. 10.26})$$


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## 608. BAYES' THEOREM VI

Here we have:

- Prior information, also called **Prior**,

$$X \sim U(0, 1), \quad \text{i.e., } X \sim \text{Beta}(1, 1)$$

- Data distribution (data density, **Likelihood**),

$$Y \mid X \sim B(n, x)$$

- Information after observation, also called **Posterior**,

$$X \mid Y \sim \text{Beta}(y+1, n-y+1)$$


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## 609. CONDITIONAL EXPECTATION I

**Definition 17.8**

Let  $X$  and  $Y$  be real-valued random variables. If  $f_{X|Y}(x|y)$  is the conditional density of  $X$  given  $Y = y$  with dominating measure  $\mu$ , then

$$\mathbb{E}(X | Y = y) = \int x f_{X|Y}(x, y) d\mu(x)$$

is the conditional expectation of  $X$  given  $Y = y$ .

Thus the conditional expectation of  $X$  given  $Y$

$$\mathbb{E}(X | Y) : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{with } \mathbb{E}(X | Y)(y) := \mathbb{E}(X | Y = y),$$

is a random variable.

## 610. CONDITIONAL EXPECTATION II

**Example 17.9 (Continuation of Example 17.2)**

We have

$$X | Y \sim B(Y, \tfrac{1}{2}),$$

so

$$\mathbb{E}(X | Y) = \tfrac{1}{2} \cdot Y.$$

## 611. LAW OF ITERATED EXPECTATION I

**Theorem 17.10 (Law of Iterated Expectation)**

$$\mathbb{E}(\mathbb{E}(X | Y)) = \mathbb{E}(X).$$

**From stack exchange:** Suppose for example that  $X, Y$  are two (positively) correlated variables, say the weight and height of persons from a given population. The expectation of the weight  $\mathbb{E}(X)$  would be my best guess of the weight of a unknown person: I'd bet for this value, if not given more data (my uninformed bet is constant). Instead, if I know the height, I'd bet for  $\mathbb{E}(X|Y)$  : that means that for different persons I'd bet a different value, and my informed bet would not be constant: sometimes I'd bet more than the "uninformed bet"  $\mathbb{E}(X)$  (for tall persons), sometime less. The natural question arises, can I say something about my informed bet in average? Well, the tower property answers: In average, you'll bet the same.

**Example: Tower Property with Dice**

Suppose we roll a fair die once. Let

$$X = \text{the die outcome}, \quad Y = \begin{cases} 0 & \text{if outcome is odd,} \\ 1 & \text{if outcome is even.} \end{cases}$$

So  $Y$  tells us whether the die result is odd or even.

$$\mathbb{E}(X) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Compute the conditional expectations.

If  $Y = 0$  (odd result), then  $X \in 1, 3, 5$  equally likely:

$$\mathbb{E}(X | Y = 0) = \frac{1 + 3 + 5}{3} = 3$$

If  $Y = 1$  (even result), then  $X \in 2, 4, 6$  equally likely:

$$\mathbb{E}(X | Y = 1) = \frac{2 + 4 + 6}{3} = 4$$

Thus:

$$\mathbb{E}(X | Y) = \begin{cases} 3 & \text{if } Y = 0, \\ 4 & \text{if } Y = 1. \end{cases}$$

Since  $Y = 0$  and  $Y = 1$  are equally likely (probability  $\frac{1}{2}$  each):

$$\mathbb{E}(\mathbb{E}(X | Y)) = \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 4 = 3.5$$

$$\mathbb{E}(\mathbb{E}(X | Y)) = \mathbb{E}(X)$$

#### 614. LAW OF TOTAL VARIANCE

**Theorem 17.12 (Law of Total Variance)**

Let  $X|Y$  be a continuous conditional random variable. Then:

$$\mathbb{V}(X) = \mathbb{E}(\mathbb{V}(X|Y)) + \mathbb{V}(\mathbb{E}(X|Y)).$$

#### 615. EXAMPLE I

**Example 0.16** (Example 17.13). Let  $X$  and  $Y$  be random variables with joint density

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{x}, & \text{for } 0 < y \leq x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_{X,Y}(x, y) = \frac{1}{x} \mathbf{1}_{(0,1)}(x) \mathbf{1}_{(0,x]}(y) = \frac{1}{x} \mathbf{1}_{(0,1)}(y) \mathbf{1}_{(y,1]}(x).$$

#### 616. EXAMPLE II

The density of the marginal distribution of  $X$  is

$$f_X(x) = \int_0^x \frac{1}{x} dy = \mathbf{1}_{(0,1)}(x).$$

The density of the marginal distribution of  $Y$  is

$$f_Y(y) = \int_y^1 \frac{1}{x} dx = \log\left(\frac{1}{y}\right) \mathbf{1}_{(0,1)}(y).$$

#### 617. EXAMPLE III

The density of the conditional distribution of  $Y|X$  is

$$f_{Y|X}(y|x) = \frac{1}{x} \mathbf{1}_{(0,x)}(y).$$

The density of the conditional distribution of  $X|Y$  is

$$f_{X|Y}(x|y) = \frac{1/x}{\log(1/y)} \mathbf{1}_{[y,1)}(x).$$

$$X \sim U(0, 1) \implies \mathbb{E}(X) = \frac{1}{2}, \quad \mathbb{V}(X) = \frac{1}{12}.$$

But: what is the distribution of  $Y$ ?

## 618. EXAMPLE IV

We know that  $Y|X \sim U(0, x)$ : The expectation of a uniform  $U(0, x)$  is simply

$$\mathbb{E}(Y|X = x) = \frac{0 + x}{2} = \frac{x}{2}.$$

So we have

$$\mathbb{E}(Y|X) = \frac{X}{2}.$$

$$\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}(Y|X)) = \mathbb{E}\left(\frac{X}{2}\right) = \frac{1}{4}$$

$$\mathbb{V}(Y) = \mathbb{E}(\mathbb{V}(Y|X)) + \mathbb{V}(\mathbb{E}(Y|X)) = \frac{1}{12}\mathbb{E}(X^2) + \frac{1}{4}\mathbb{V}(X) = \frac{7}{144}$$

$$\mathbb{E}(X \cdot Y) = \mathbb{E}(\mathbb{E}(X \cdot Y|X)) = \mathbb{E}(X \mathbb{E}(Y|X))$$

$$= \mathbb{E}\left(X \cdot \frac{X}{2}\right) = \frac{1}{2}\mathbb{E}(X^2)$$

$$= \frac{1}{2}(\mathbb{V}(X) + \mathbb{E}(X)^2) = \frac{1}{2}\left(\frac{1}{12} + \frac{1}{4}\right) = \frac{1}{6}$$

$$\text{Cov}(X, Y) = \frac{1}{6} - \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{24}$$

## 619. CONDITIONAL INDEPENDENCE I

**Definition 17.14** Let  $X, Y$  and  $Z$  be random variables such that

$$f_{X,Y|Z}(x, y|z) = f_{X|Z}(x|z)f_{Y|Z}(y|z),$$

then we write  $X \perp Y | Z$ : that is,  $X$  and  $Y$  are conditionally independent given  $Z$ .

**Example 17.15** We roll three dice (red, green, and blue). Let

- $X$ : the sum of the red and green dice,
- $Y$ : the sum of the red and blue dice,
- $Z$ : the number shown on the red die.

Obviously,  $X \perp Y | Z$ .

## Q1

Given a scalar  $\beta > 0$  and a continuous random variable  $X$  with probability density function (pdf)

$$f_X(x) = \beta \exp(-2\beta|x|), \quad \forall x \in \mathbb{R}.$$

**Hint:**  $\mathbb{E}[X] = 0$ .

(a) Sketch the pdf  $f_X(x)$ .

(b) Now consider the random variable  $Y = g(X) = |X|$ . Determine the pdf  $f_Y(y)$  for all  $y \in \mathbb{R}$ .

**Hint:** Pay attention to the case distinction.

$$y < 0: \quad g(x) - y = 0 \text{ has no solution} \quad \Rightarrow \quad f_Y(y) = 0,$$

$$y \geq 0: \quad g(x) - y = 0 \text{ has the solutions } x = \pm y; \quad \left| \frac{dg}{dx} \right| = |\pm 1| = 1.$$



$$\Rightarrow f_Y(y) = f_X(y) + f_X(-y) = \beta e^{-2\beta|y|} + \beta e^{-2\beta|-y|} = 2\beta e^{-2\beta y}, \quad y \geq 0,$$

and

$$f_Y(y) = \begin{cases} 2\beta e^{-2\beta y}, & y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(Note: At  $y = 0 \Leftrightarrow x = 0$ , the derivative is not defined. The value of the pdf at isolated points can always be chosen arbitrarily without changing the distribution. Thus, for  $y = 0$ , one may use the expression  $f_Y(y) = 2\beta e^{-2\beta y}$ .)

**(c) What is the relationship between  $\mathbb{E}[X^2]$  and  $\mathbb{E}[Y^2]$ ?**

$$\mathbb{E}[Y^2] = \mathbb{E}[|X|^2] = \mathbb{E}[X^2].$$

**(d) Express  $\mathbb{E}[Y^2]$  in terms of  $\mathbb{E}[Y]$  and  $\text{Var}(Y)$ .**

$$\mathbb{E}[Y^2] = \text{Var}(Y) + (\mathbb{E}[Y])^2.$$

From now on, let  $\beta = 2$ . In this case,  $Y$  is exponentially distributed with parameter  $\lambda = 4$ .

**(e) State the expectation and variance of  $Y$ .**

$$\mathbb{E}[Y] = \frac{1}{\lambda} = \frac{1}{4}, \quad \text{Var}[Y] = \frac{1}{\lambda^2} = \frac{1}{16}.$$

**(f) Compute  $\text{Var}[X]$ .**

$$\text{Var}[X] = \mathbb{E}[X^2] - 0^2 = \mathbb{E}[Y^2] = \text{Var}[Y] + (\mathbb{E}[Y])^2 = \frac{1}{16} + \left(\frac{1}{4}\right)^2 = \frac{1}{8}.$$

## Q2

A factory produces integrated circuits (ICs), which are functional with probability  $\frac{5}{6}$ .

Let  $X_i$  be a random variable that indicates whether the  $i$ -th IC is functional ( $X_i = 1$ ) or not ( $X_i = 0$ ). Assume that the  $X_i$  are stochastically independent.

**(a) Give the expectation of  $X_i$ .**

$$\mathbb{E}[X_i] = \frac{5}{6} \quad (\text{since } X_i \text{ is Bernoulli distributed with parameter } p = \frac{5}{6}).$$

**(b) Give the variance of  $X_i$ .**

$$\text{Var}[X_i] = \frac{5}{6} \left(1 - \frac{5}{6}\right) = \frac{5}{36}$$

(since  $X_i$  is Bernoulli distributed with parameter  $p = \frac{5}{6}$ ).

**(c) Let**

$$Y = \sum_{i=1}^N X_i.$$

**What type of distribution does  $Y$  have?**

$Y$  is binomially distributed.

**(d) Give the expectation of  $Y$  as a function of  $N$ . Provide a calculation or other justification.**

$$\mathbb{E}[Y] = \mathbb{E}\left[\sum_{i=1}^N X_i\right] = \sum_{i=1}^N \mathbb{E}[X_i] = \frac{5}{6} N.$$

(Alternative: since  $Y \sim \text{Binomial}(N, \frac{5}{6})$ , its mean is  $Np$ .)

Let  $Z = \frac{1}{N} \sum_{i=1}^N X_i$ .

(e) **Compute  $\text{Var}(Z)$  as a function of  $N$ . Indicate the step where you use independence.**

$$\text{Var}(Z) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) = \frac{1}{N^2} \text{Var}\left(\sum_{i=1}^N X_i\right) \stackrel{\text{indep.}}{=} \frac{1}{N^2} \sum_{i=1}^N \text{Var}(X_i) = \frac{1}{N^2} \cdot N \cdot \frac{5}{36} = \frac{5}{36} \frac{1}{N}.$$


---

### Q3

Given the following probability density function of the random variable  $X$ :

$$f_X(x) = \begin{cases} 2 - 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- Compute the cumulative distribution function  $F_X(x)$  of  $X$ .
  - Sketch  $f_X(x)$  and  $F_X(x)$  in separate coordinate systems and label the axes.
  - Compute the probability  $\mathbb{P}(A)$  of the event  $A = \{\frac{1}{2} \leq X \leq 1\}$ .
- 

### Q4

Given the following probability density function of the random variable  $X$ :

$$f_X(x) = \begin{cases} x + c, & |x| \leq a, \\ 0, & \text{otherwise,} \end{cases}$$

with parameters  $a, c \in \mathbb{R}^+$ .

- What conditions must the parameters  $a$  and  $c$  satisfy so that  $f_X(x)$  is a valid probability density function?
  - Determine  $a$  as a function of  $c$  and give a numerical condition for  $c$ .
  - Give one explicit example for  $f_X(x)$ .
- 

### Q5

Let  $Y$  be a discrete random variable with probability mass function

$$(3) \quad p_Y(y) = \begin{cases} \frac{y(b-y)}{20}, & y \in \{1, 2, 3, 4\}, \\ 0, & \text{otherwise,} \end{cases}$$

- How must  $b$  be chosen so that  $p_Y(y)$  is a valid probability mass function?  
For the  $b$  you found:
  - Sketch  $p_Y(y)$  and  $F_Y(y)$  in separate coordinate systems and label the axes.
  - What are the values of  $p_Y(\frac{5}{2})$  and  $F_Y(\frac{5}{2})$ ?
-

## Q6

Given the following probability density function of the random variable  $X$ :

$$(1) \quad f_X(x) = \begin{cases} |x|, & x \in [-1, 1], \\ 0, & \text{otherwise,} \end{cases}$$

Using the following mapping, define the random variable  $Y = g(X)$ :

$$(2) \quad g(X) = -3|X|.$$

- (a) Onto which range  $\Omega_Y$  is  $\Omega_X = [-1, 1]$  mapped? What value does  $f_Y(y)$  take for  $y \notin \Omega_Y$ ?
  - (b) Determine the roots/solutions  $x_i$  of  $y - g(x_i) = 0$  for  $y \in \Omega_Y$ ,  $i \in \mathbb{N}$ . (Take into account the piecewise definition of  $g(X)$ .)
  - (c) Using this, give the pdf  $f_Y(y)$ .
- 

## Q7

Let  $X$  be a uniformly distributed random variable on  $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ .

- (a) Give the pdf  $f_X(x)$ .

Now  $X$  is transformed into a new random variable  $Y$  via the mapping

$$Y = g(X) = \sin(X).$$

- (b) On which interval is the pdf  $f_Y(y)$  of  $Y$  (according to the mapping  $Y = g(X)$ ) nonzero?
  - (c) Compute  $f_Y(y)$  for all  $y \in \mathbb{R}$ . **Hint:**  $\cos(x) = \sqrt{1 - \sin^2(x)}$ .
  - (d) Give the cdf  $F_Y(y)$ . **Hint:**  $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x)$ .
- 

## Q8

Given a random variable  $X$  with probability density function (pdf)

$$f_X(x) = \begin{cases} |x|^3, & x \in [-c, c], \\ 0, & \text{otherwise,} \end{cases} \quad \text{with } c > 0.$$

- (a) Sketch  $f_X(x)$  qualitatively for  $x \in \mathbb{R}$ .

Now define the random variable  $Y = X^2$ .

- (c) Give the smallest possible interval  $\Omega_Y$  (as a function of  $c$ ) that contains all possible realizations of  $Y$ .
  - (d) Determine the probability density function (pdf)  $f_Y(y)$  for  $y \in \Omega_Y$ .
- 

## Q9

In the following problem we study the time until the next time a professional footballer of FC Barcelona gets injured. This is modeled by the random variable  $T$  with the probability density function  $f_T(t)$  given below. We assume that at  $t = 0$  no one is injured yet.

$$(1) \quad f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

with  $t$  measured in days and  $\lambda = \frac{1}{28}$ . Throughout, take 1 month =  $m = 30$  days and 1 week =  $w = 7$  days.

- (a) According to the standard stochastic models from the lecture, what distribution does  $T$  have?

(b) With what probability are the FC Barcelona players still uninjured after 5 weeks?

**Hint:**  $\sqrt[4]{e^5} \approx \frac{1}{e^{2.865}}$ .

(c) Suppose that after two months fortunately no player has been injured yet. What is now the probability that this remains so for the five following weeks?

Clearly,  $T$  is a continuous real-valued random variable. From this we now construct a discrete random variable that records the day on which the first player gets injured. A first naive attempt is to evaluate  $f_T(t)$  at integer times  $t = k$ :

$$(2) \quad p_X(k) = \begin{cases} c e^{-\lambda k}, & k \in \mathbb{N}_0, \\ 0, & \text{otherwise,} \end{cases}$$

where the constant  $c$  must be (re)computed since the values of  $f_T(t)$  are not probabilities.

(d) Compute this constant  $c$ .

**Hint:**  $\sum_{i=0}^{\infty} a^i = \frac{1}{1-a}$  for  $|a| < 1$ .

A more systematic variant is to set  $p_Y(k)$  equal to the probability of a specific interval in  $T$ . Thus,

$$p_Y(k) = \mathbb{P}(\{k \leq T \leq k+1\}).$$

(e) What is  $p_Y(k)$  for all  $k \in \mathbb{Z}$ ?

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